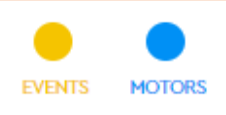
Úloha 1 – Katapult



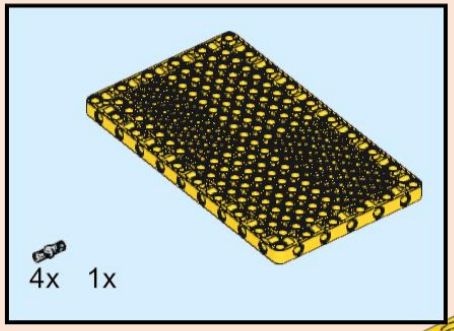
	
	60 min
Téma	Katapult, Naviják, Vrh na vzdálenost, Vrh na přesnost (Stavba a bourání zdí)

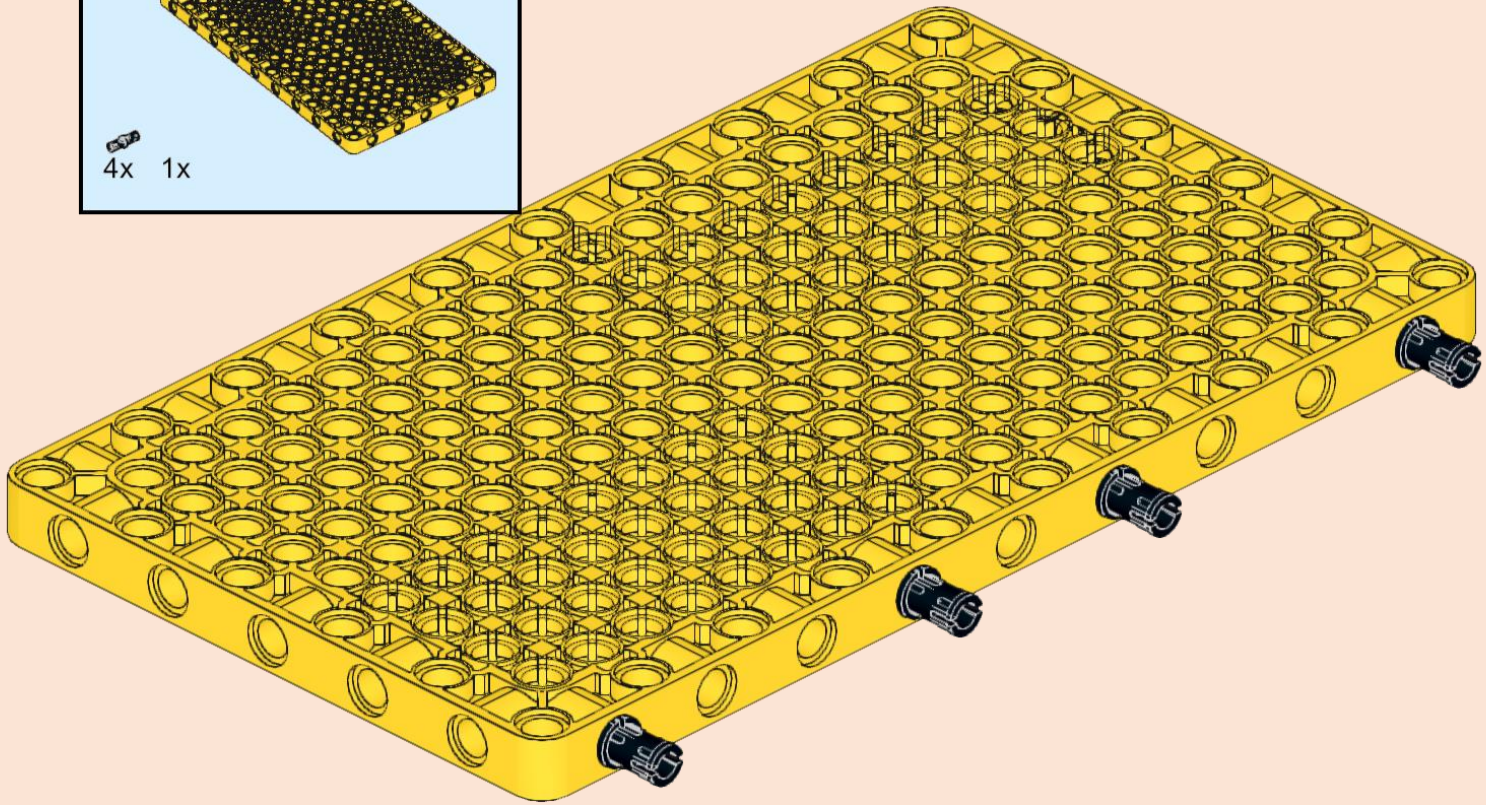


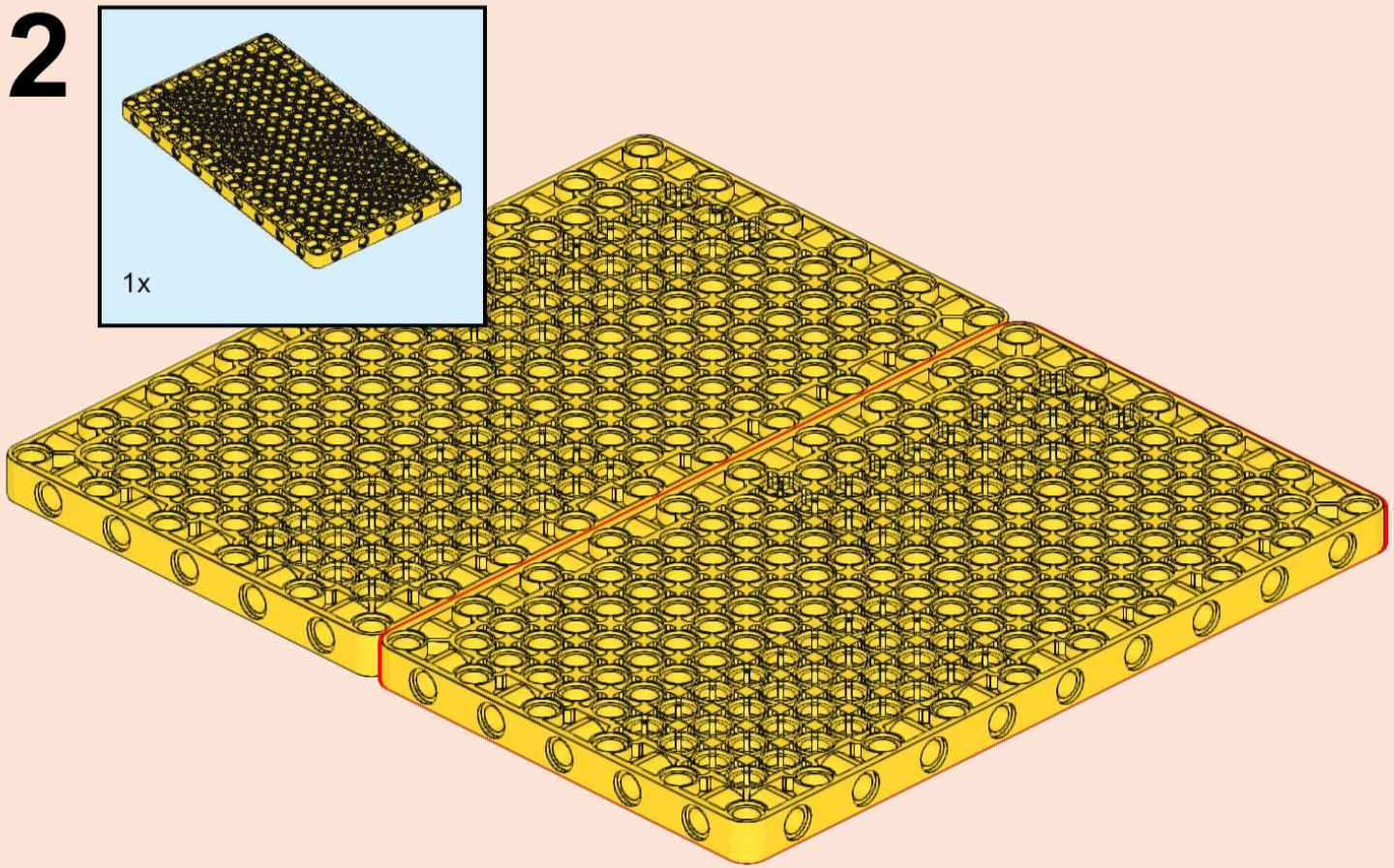
Co sestavit a připravit:

- ☐ Sestroj katapult podle návodu.
- ☐ Připrav si dostatečnou délku provázku/nitě, přivaž ji na navíjecí motor a protáhni oky/volně a na konec připevni navíjecí kolík, kterým se přitahuje konec ramena katapultu. (uvolňuje se ručně)
- ☐ Připrav si projektil, který se bude vrhat. (dobře zohýbaný papír, 3D vytisknuté „kameny“, dřevěné kostičky…)

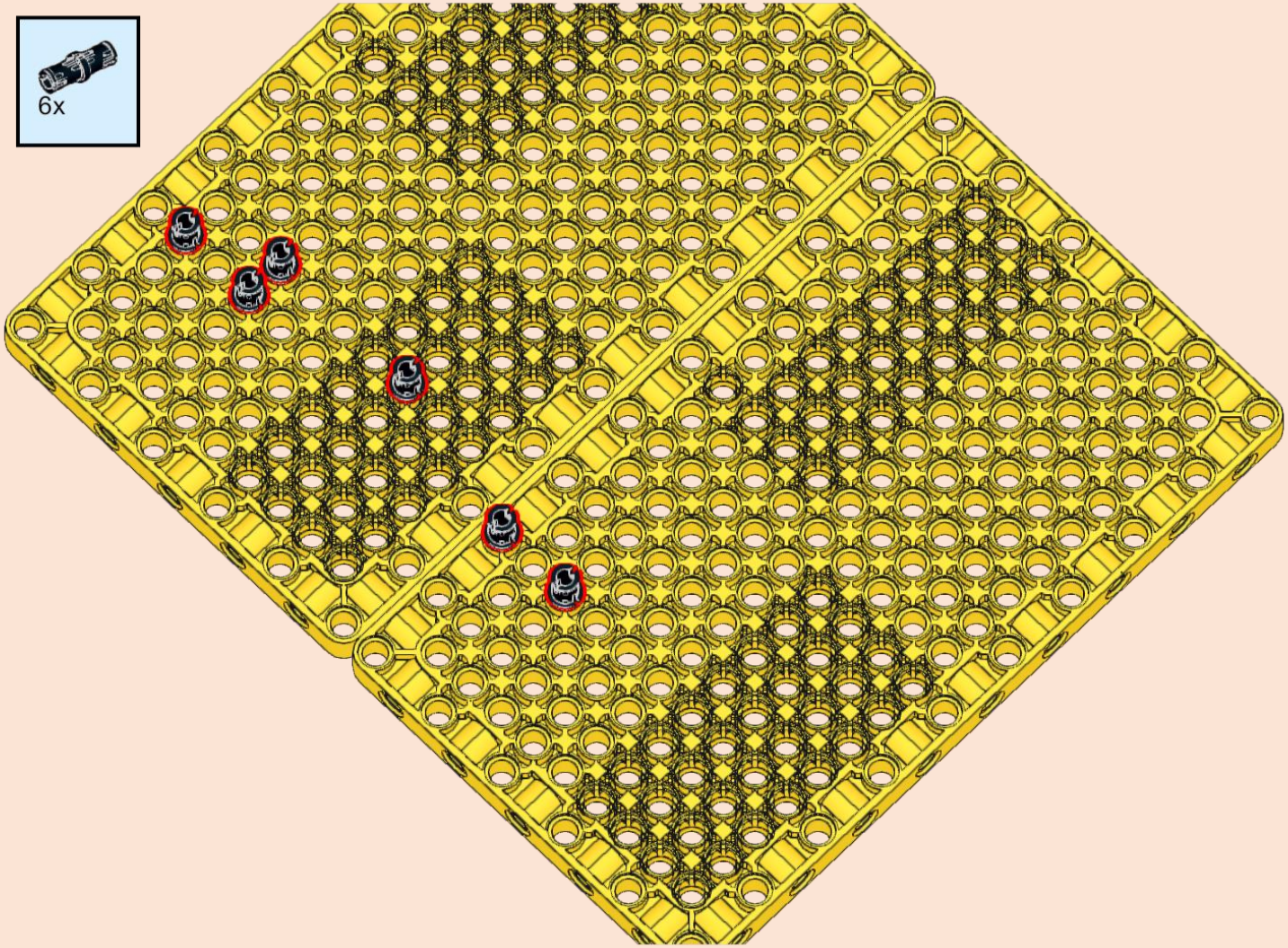
1



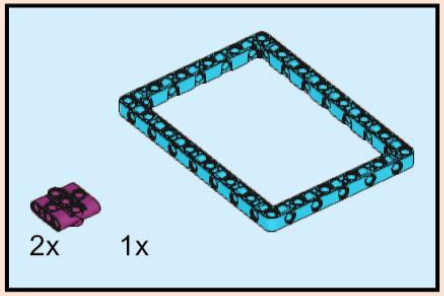


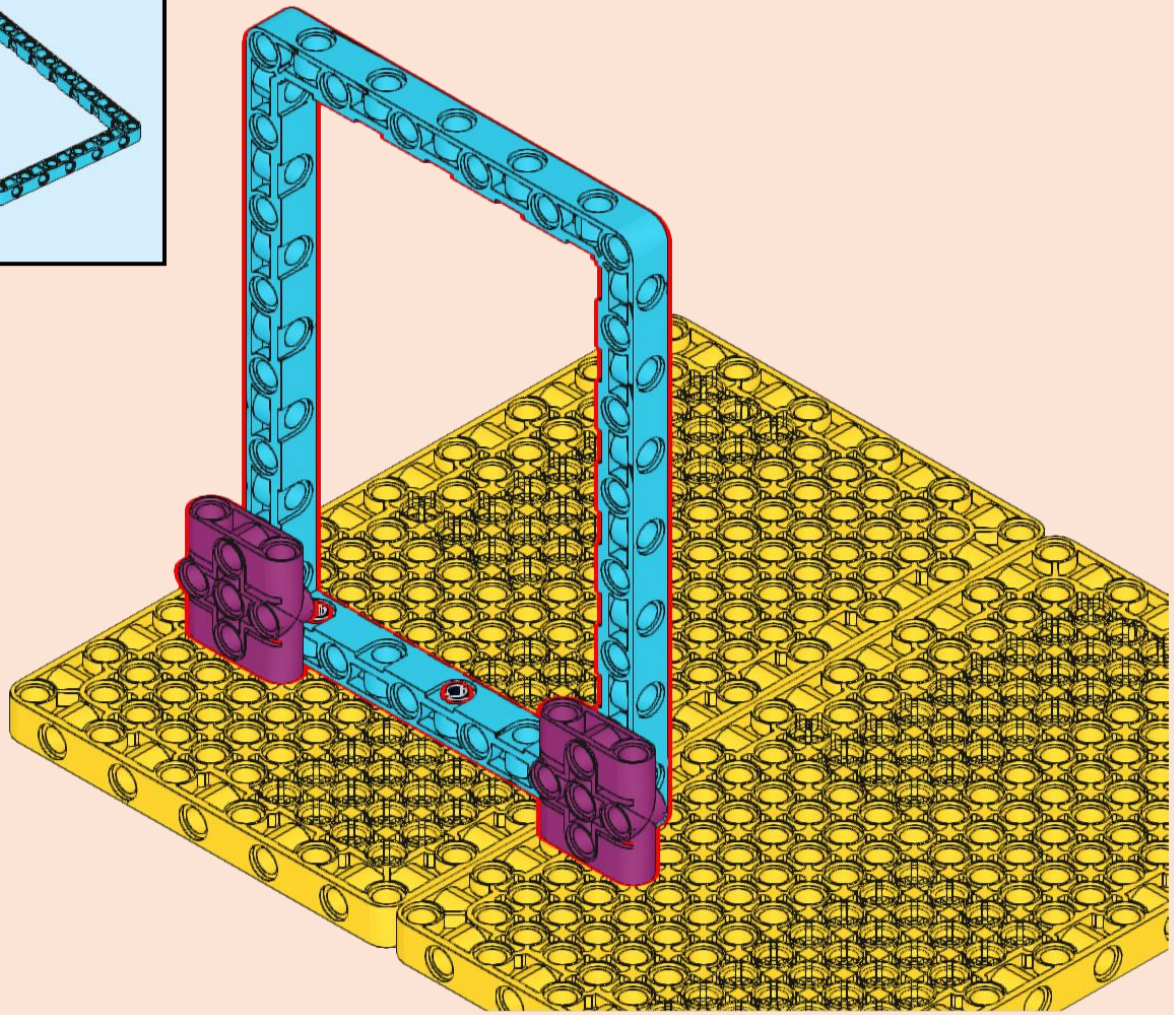

2
1x

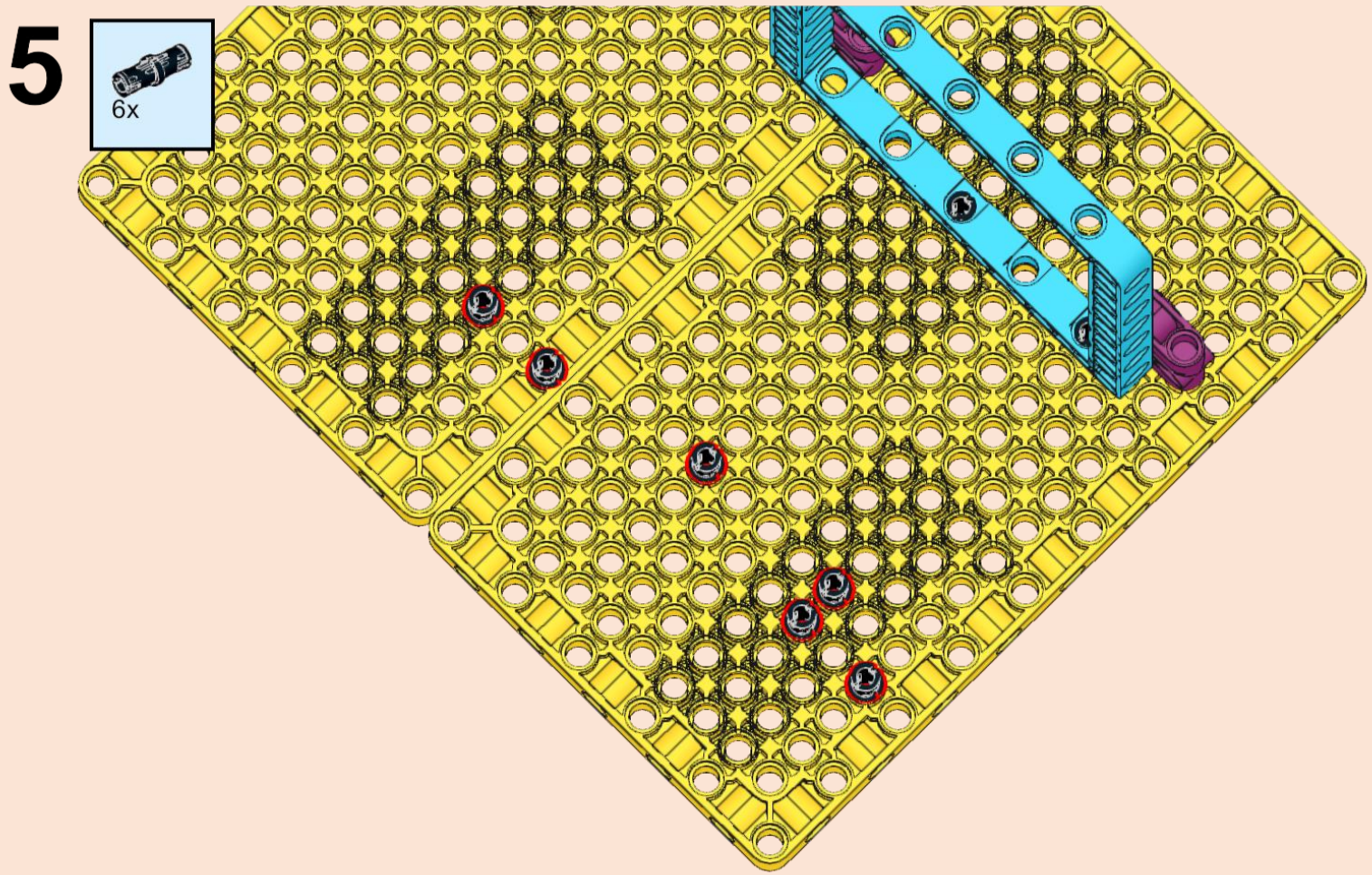
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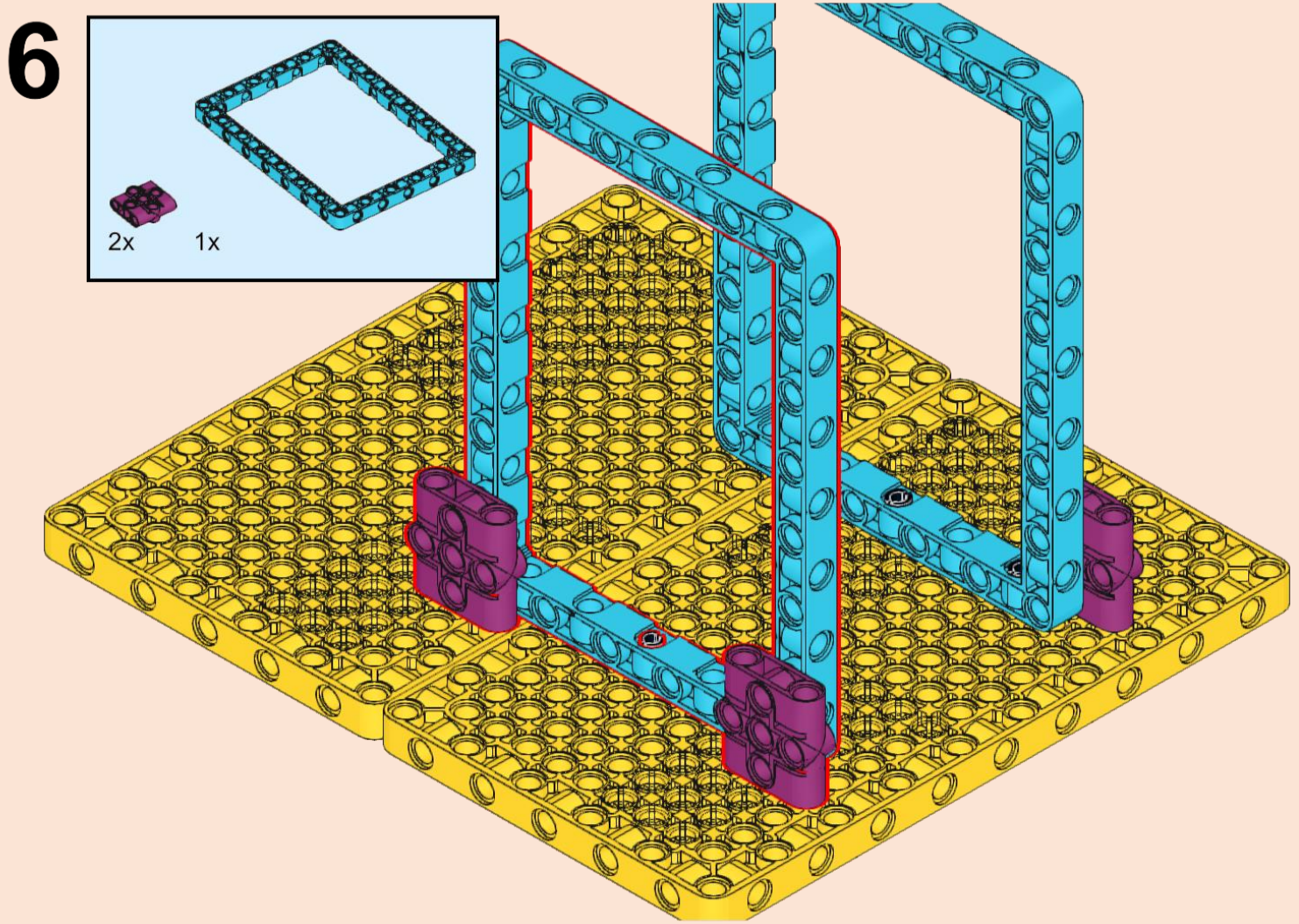


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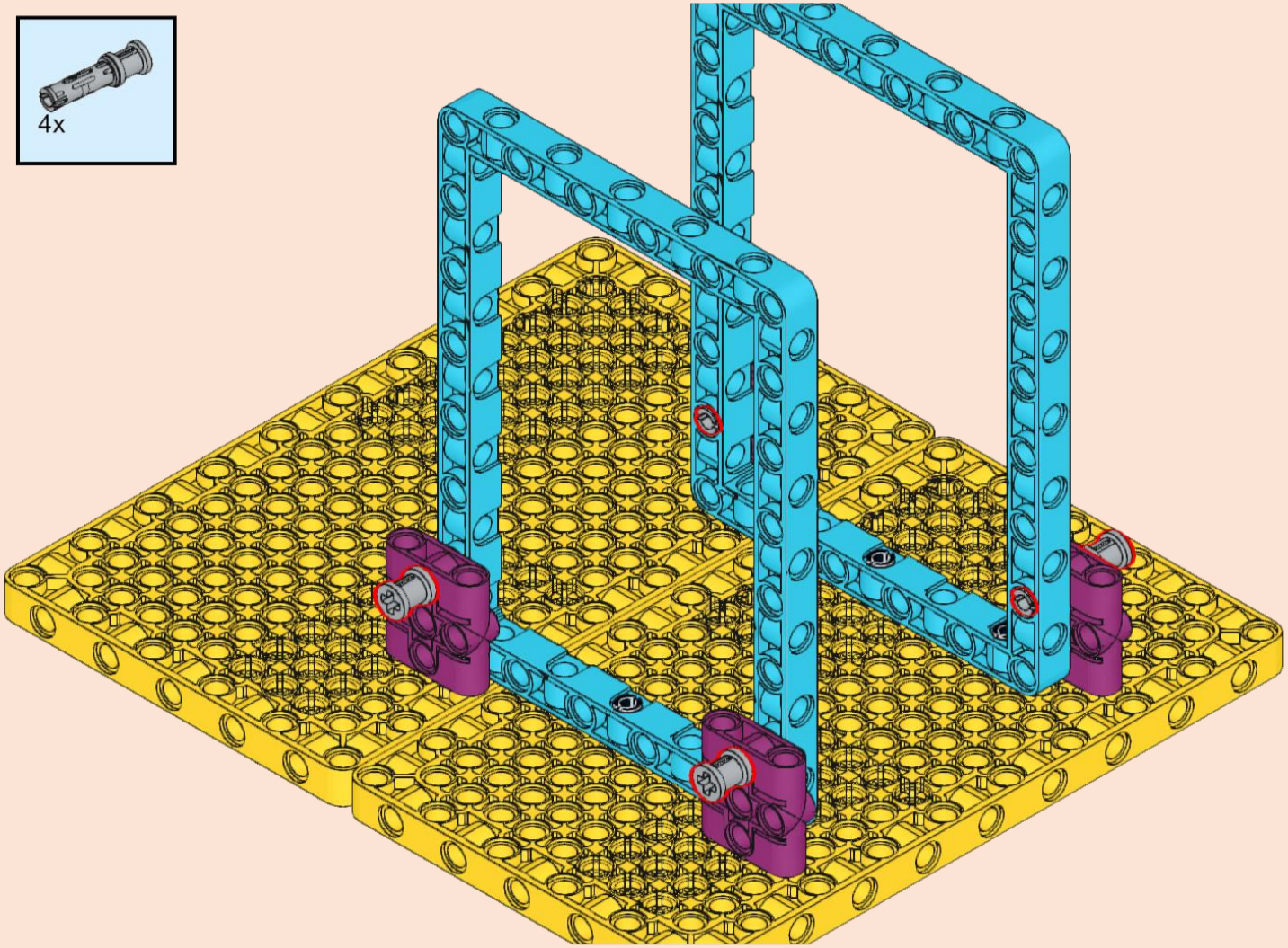




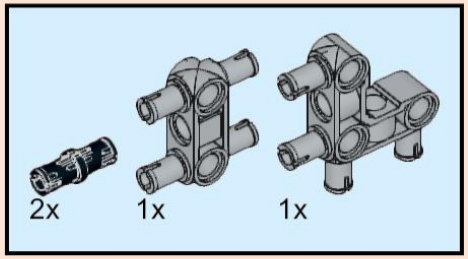

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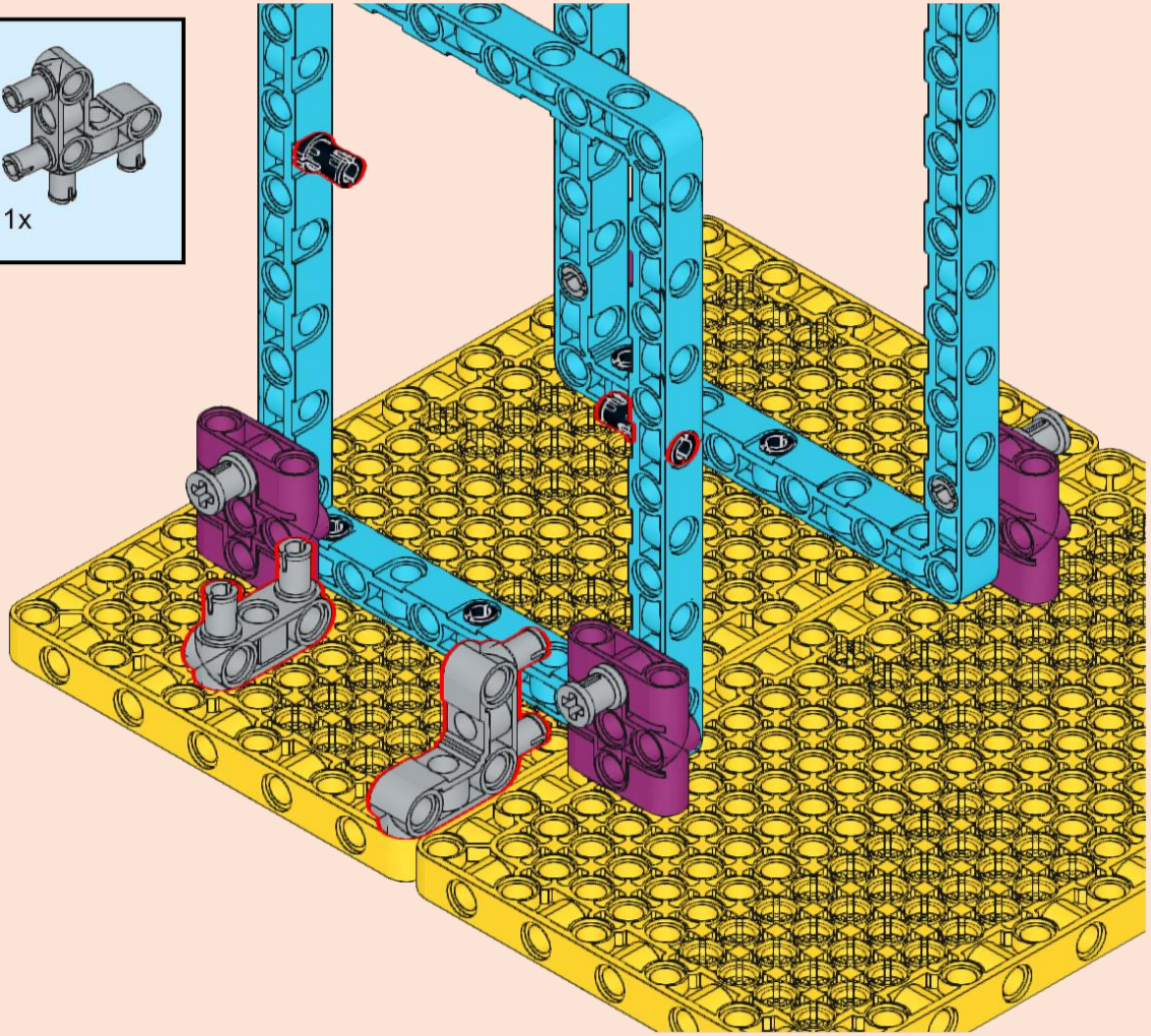

6
2x
1x

7

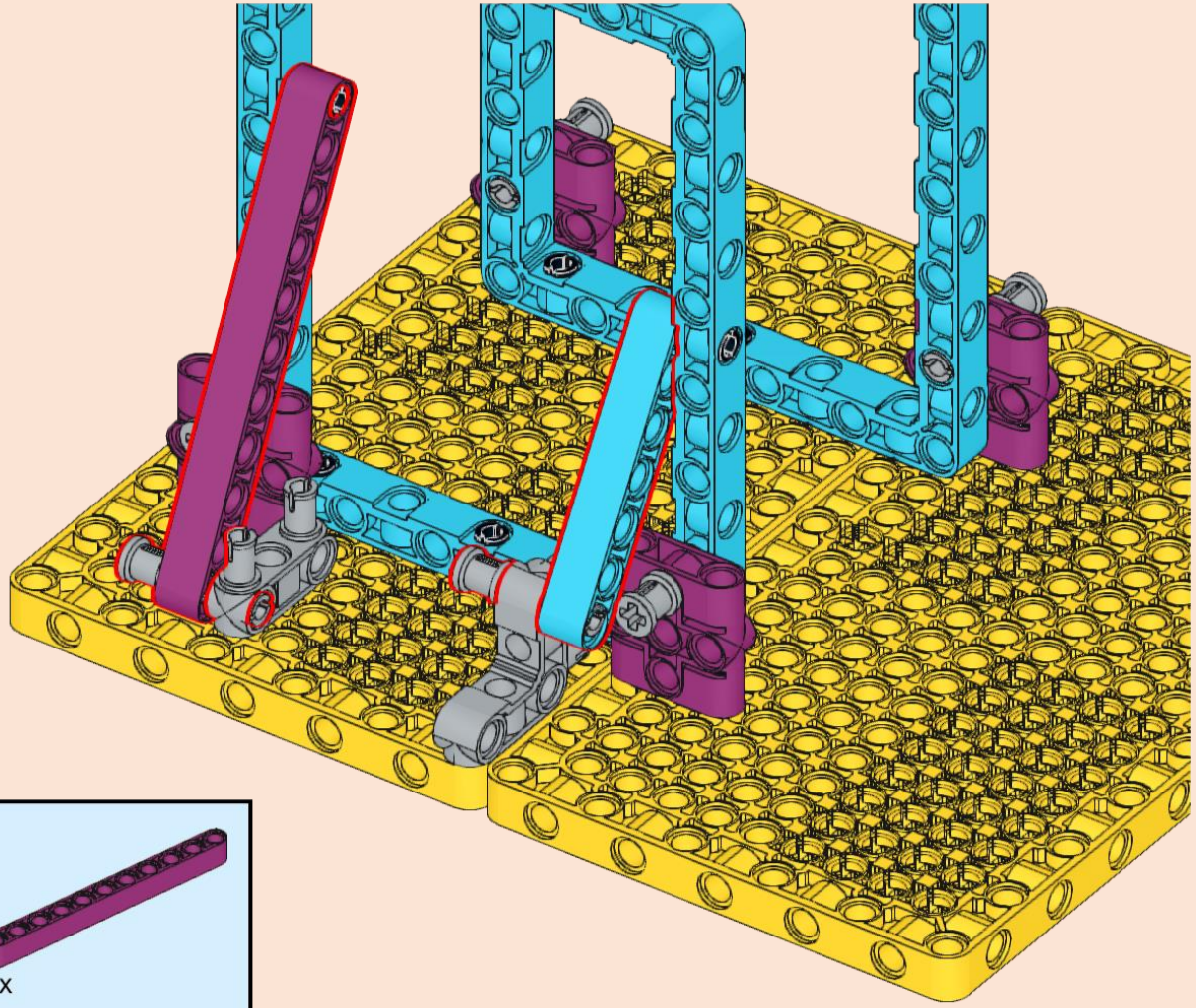


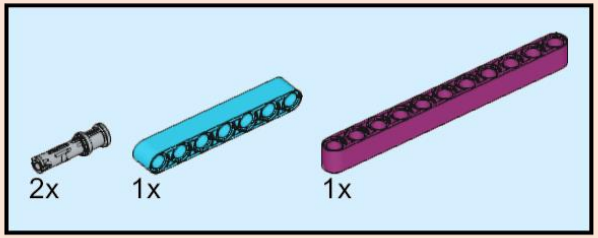
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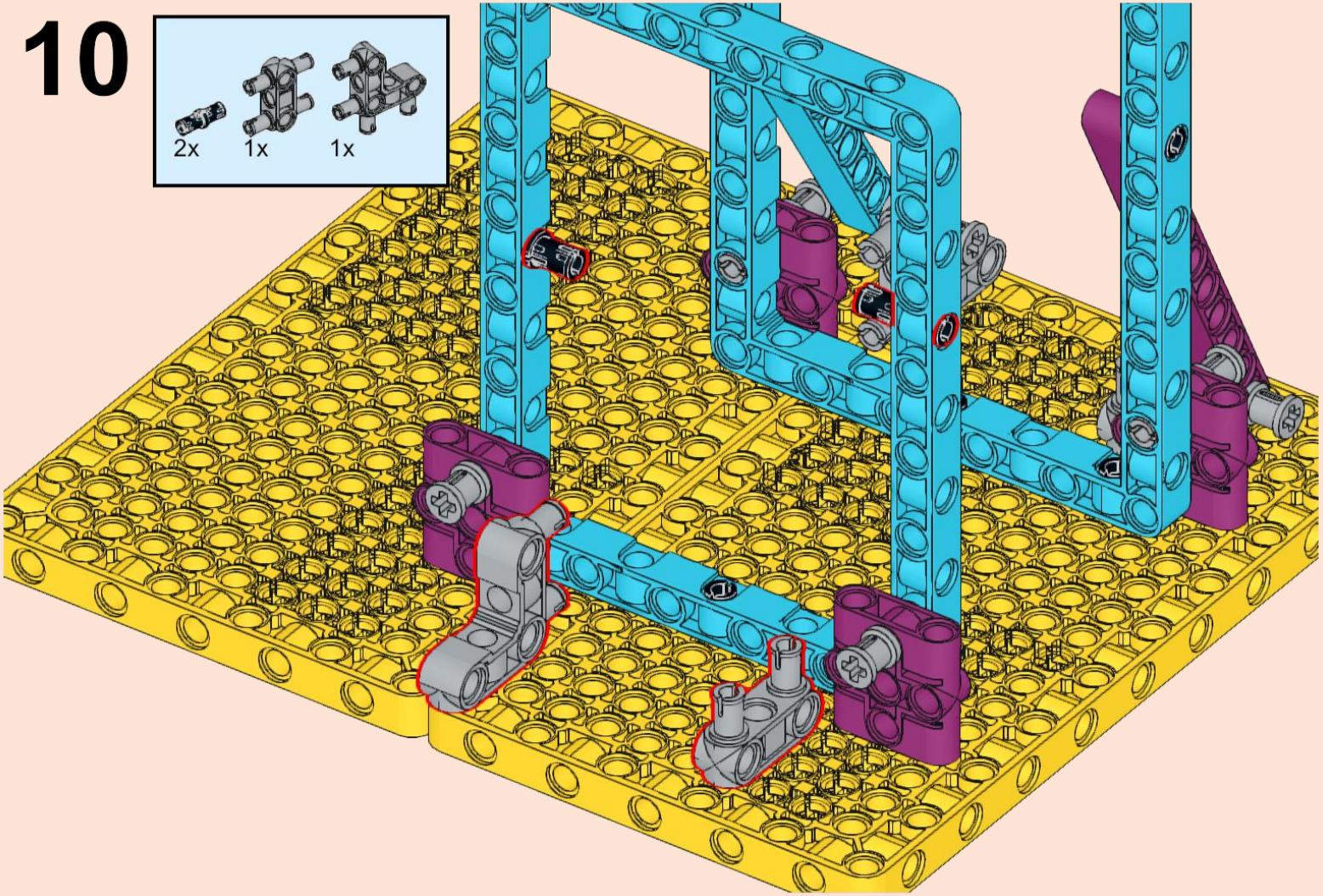




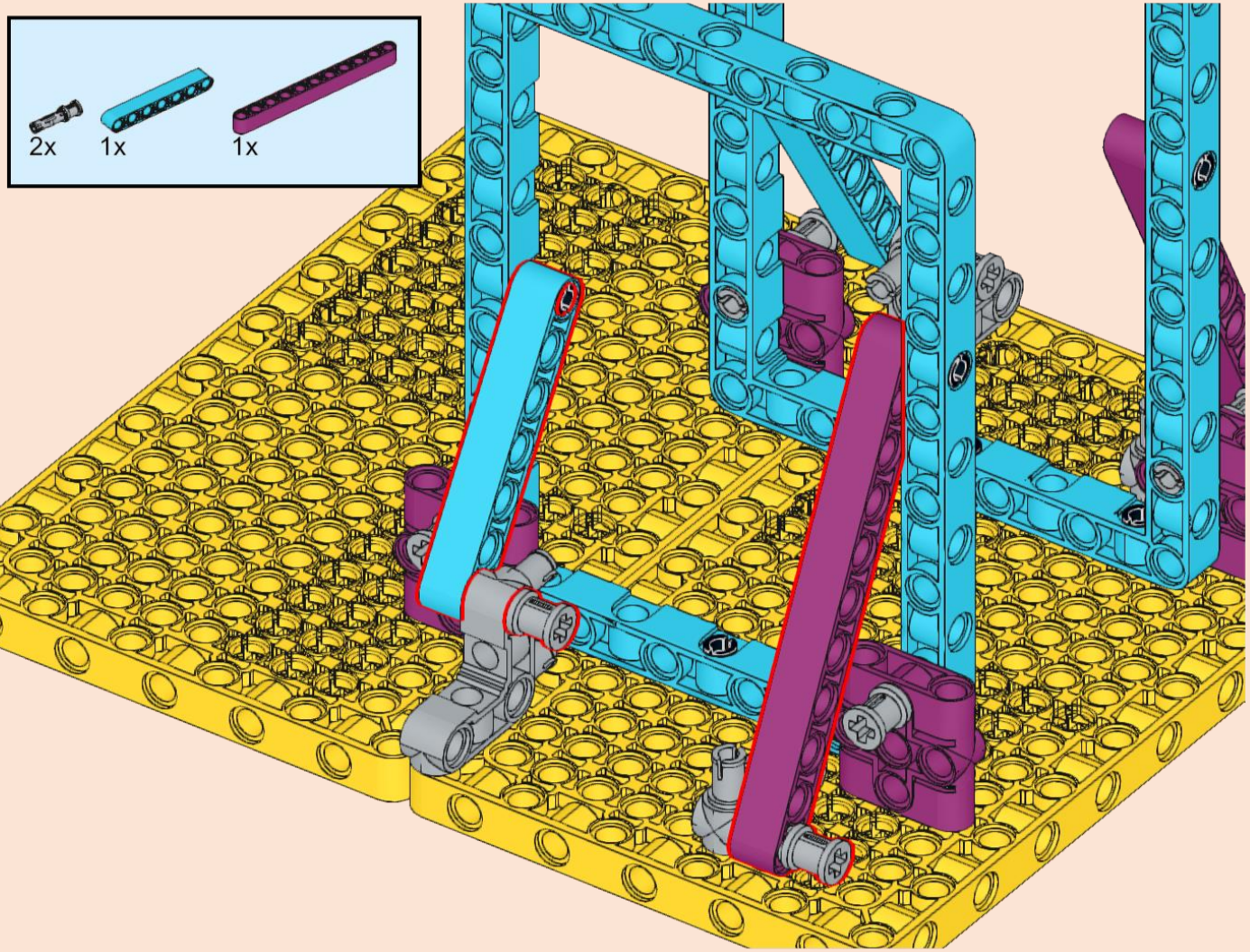
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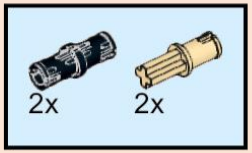


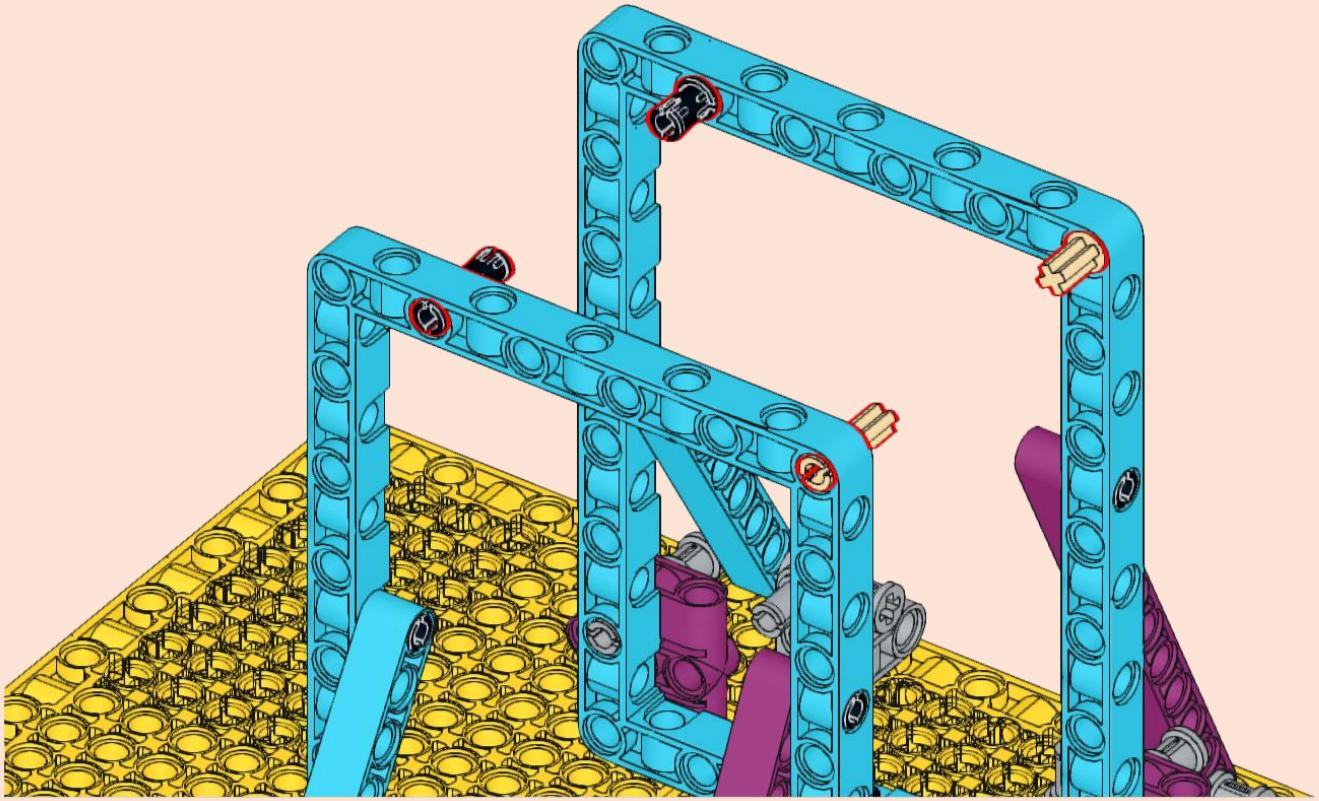

10
2x
1x
1x

11

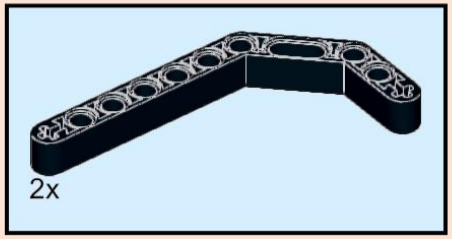


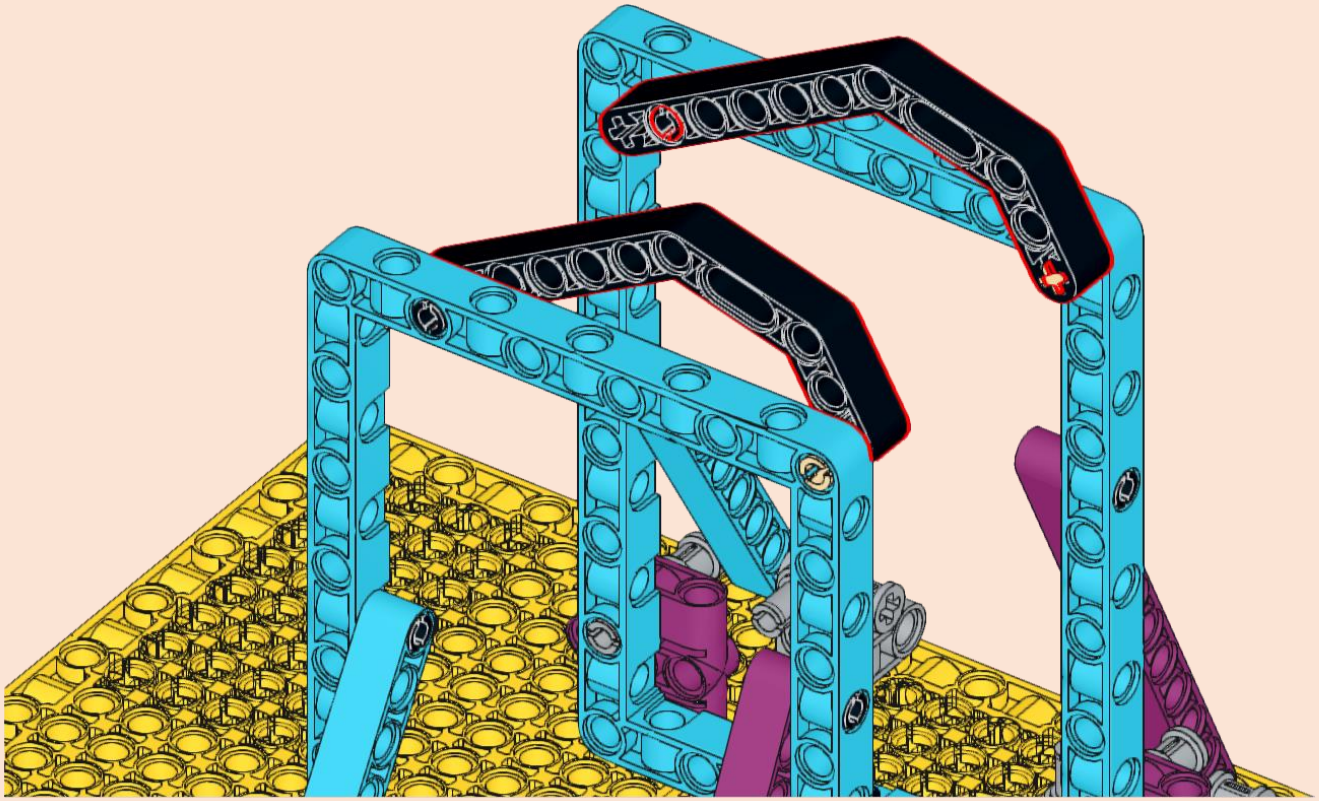
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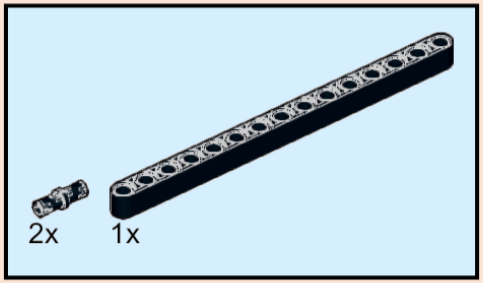


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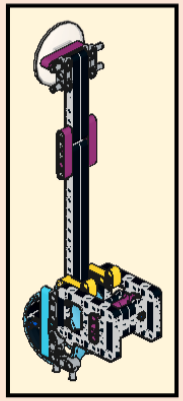




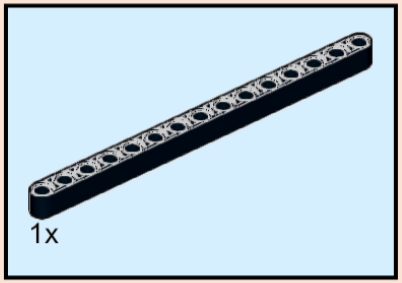
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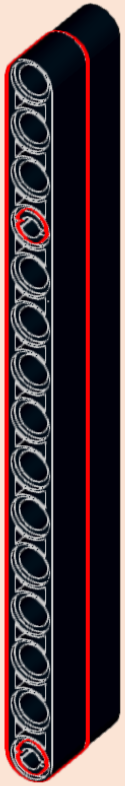




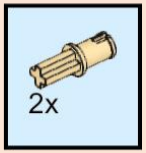


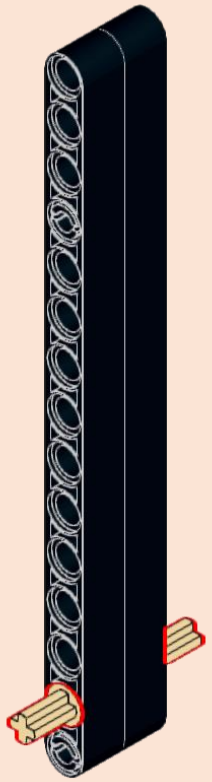
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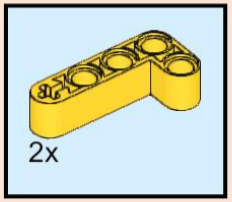


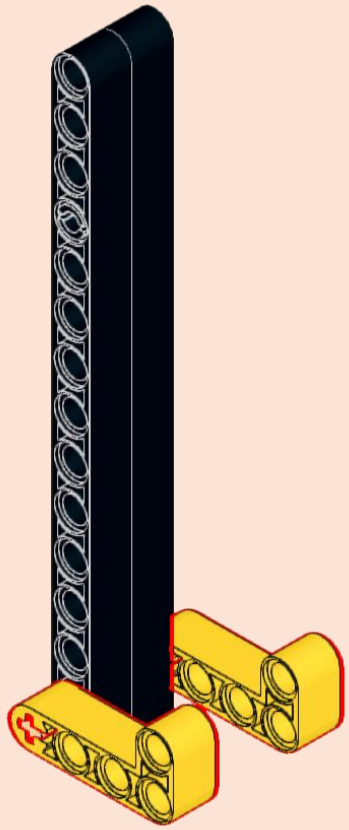
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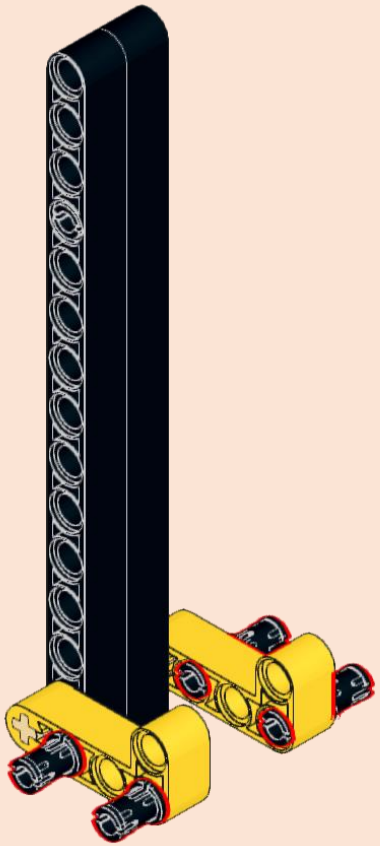
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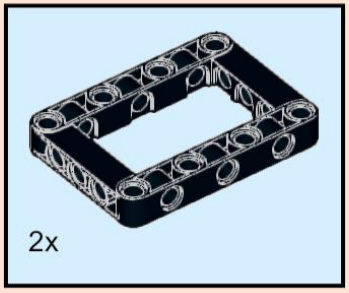


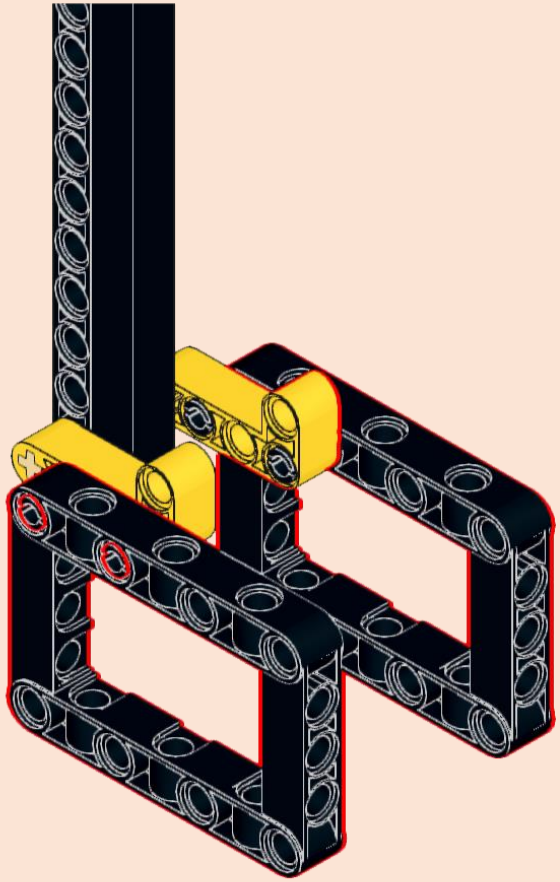
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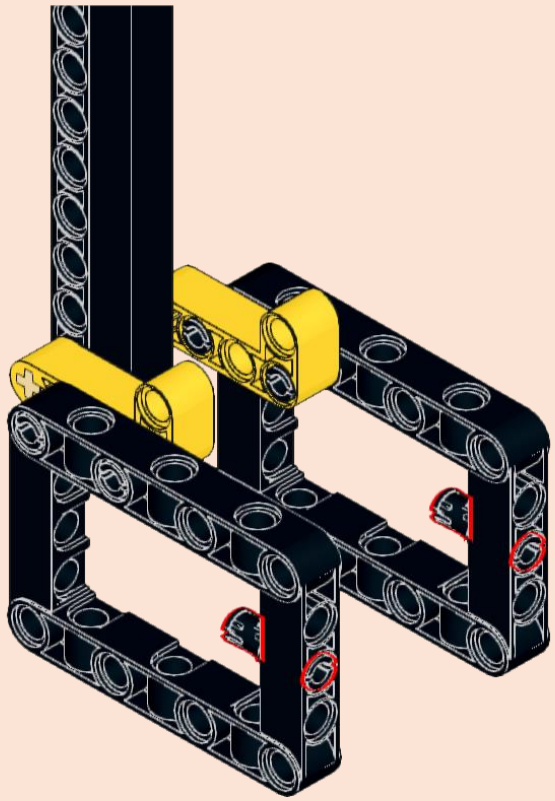
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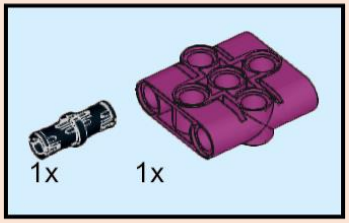


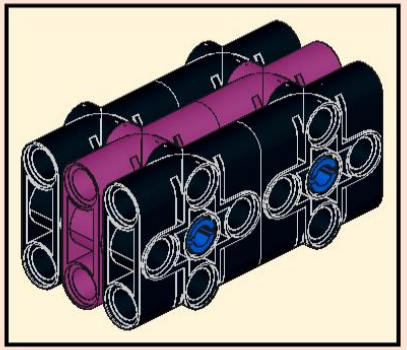
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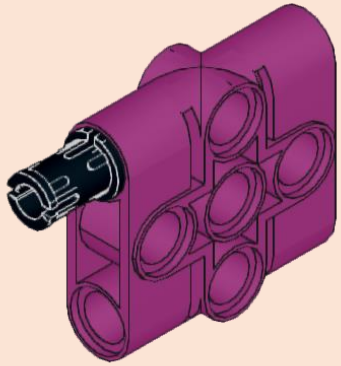




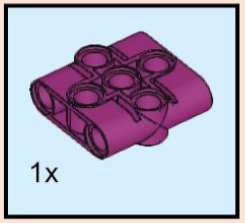
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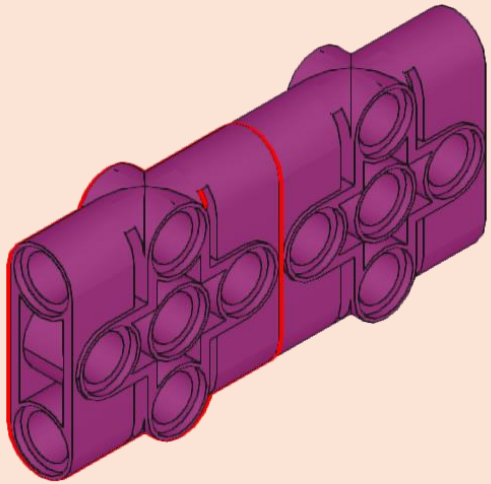






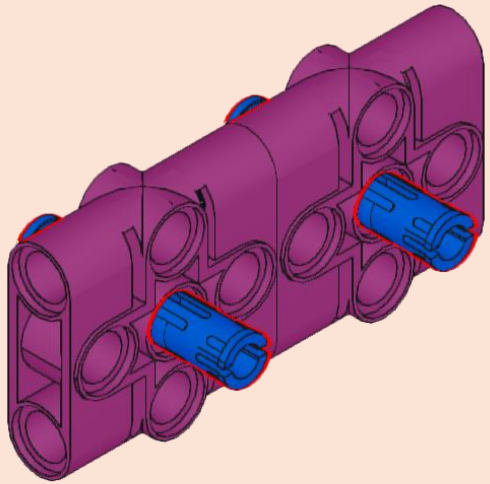
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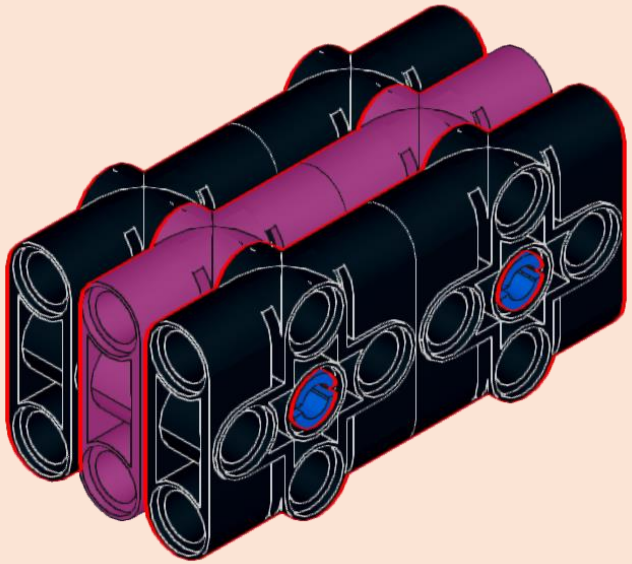
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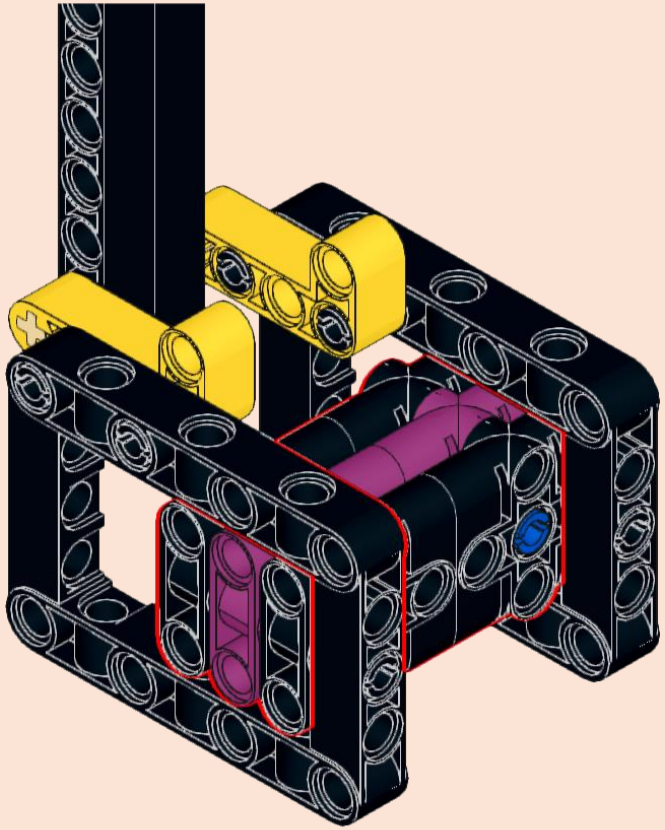


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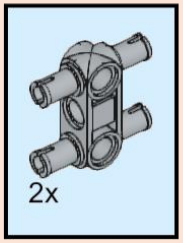


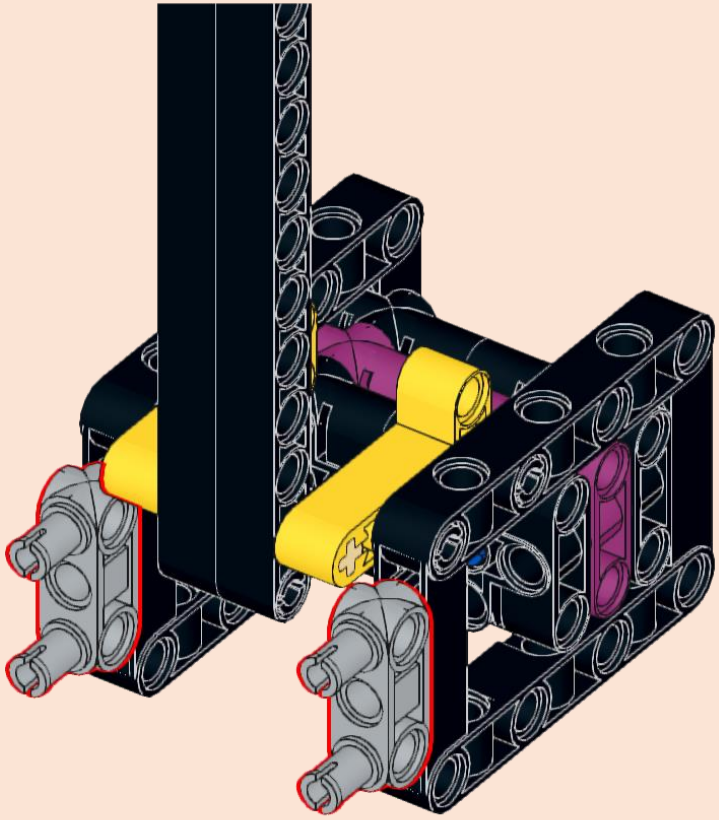


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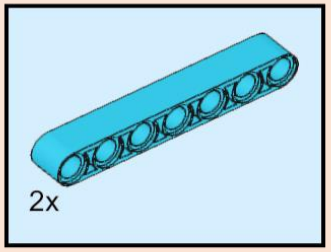


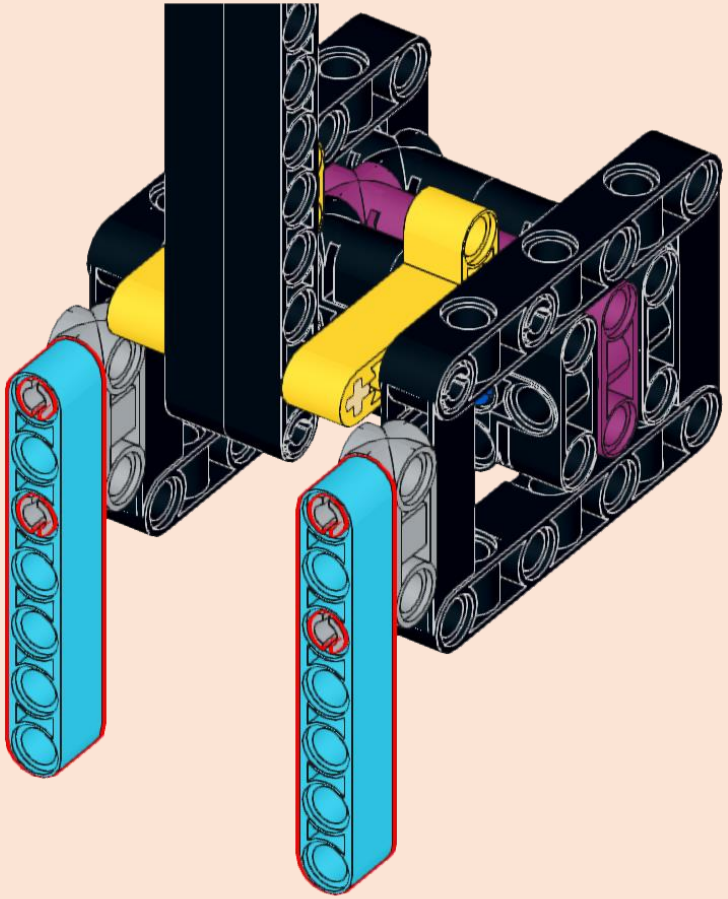
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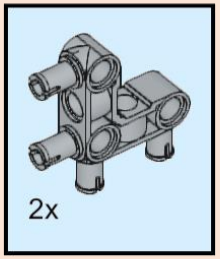


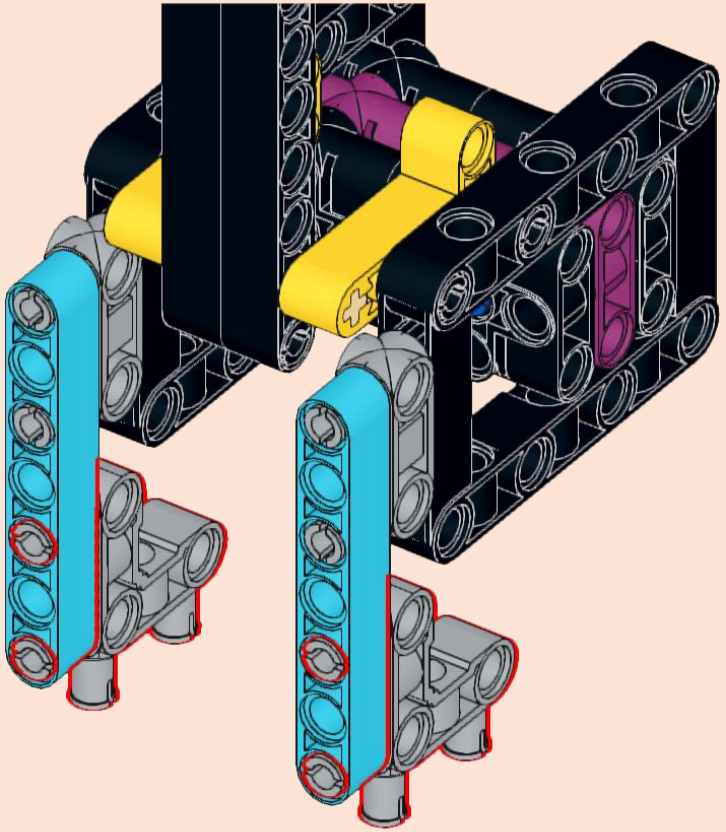
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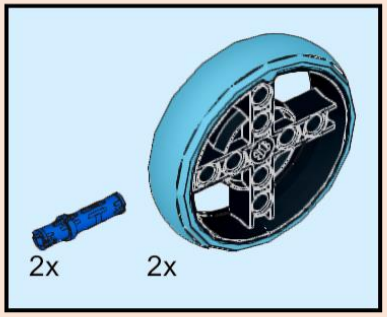


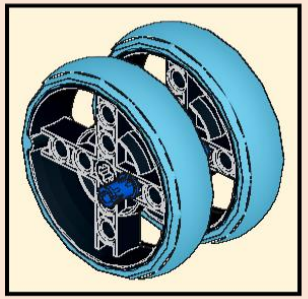
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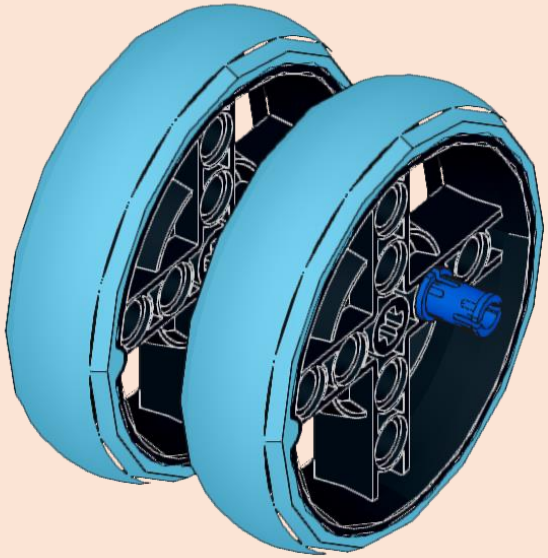




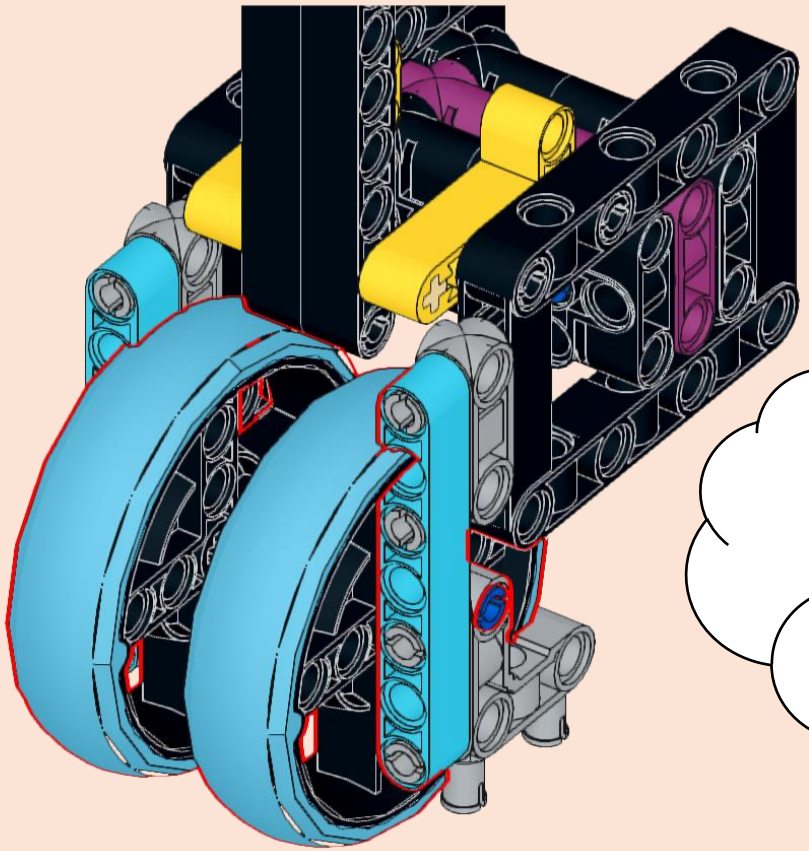
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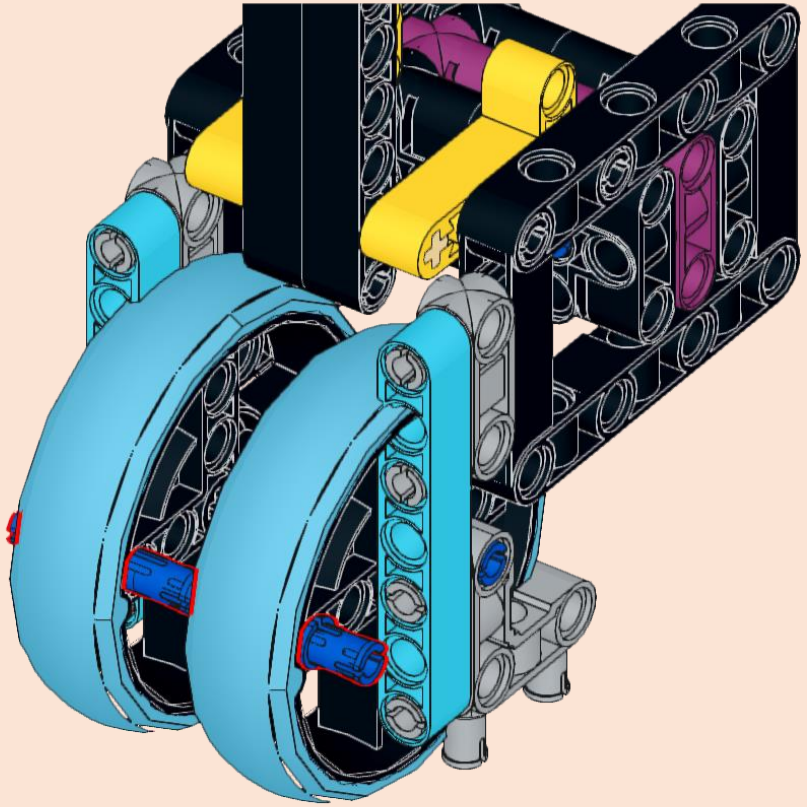
30




Ty kola možná nebudou nejpevnější závaží na světě, ale fungovat to bude!

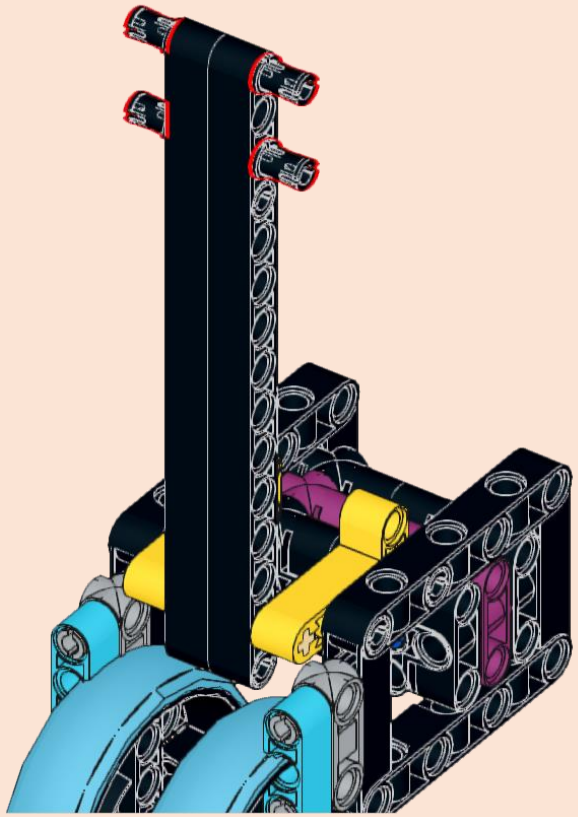
31



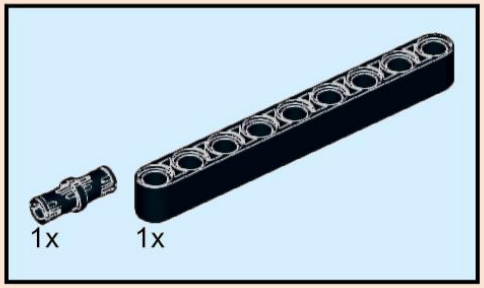


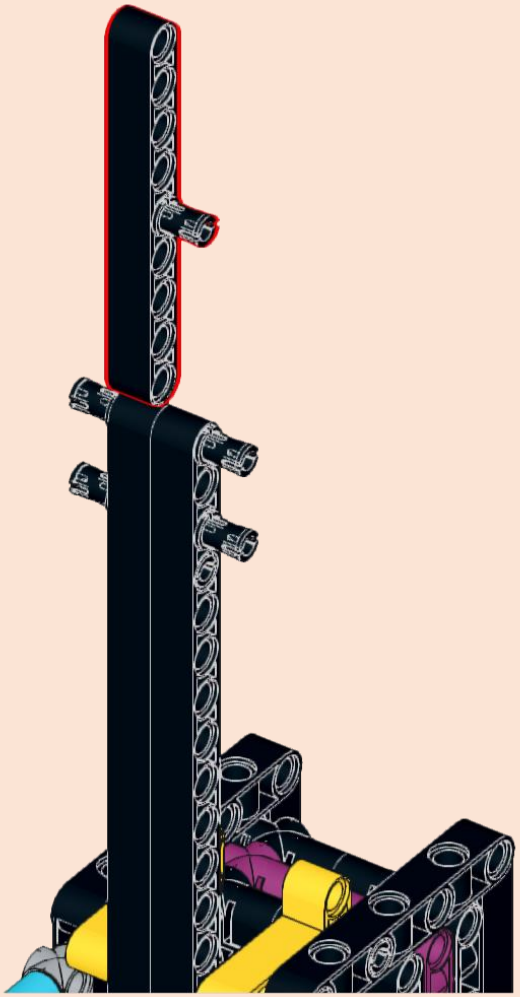
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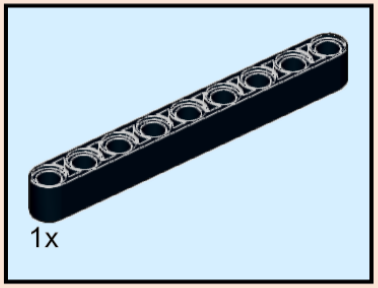


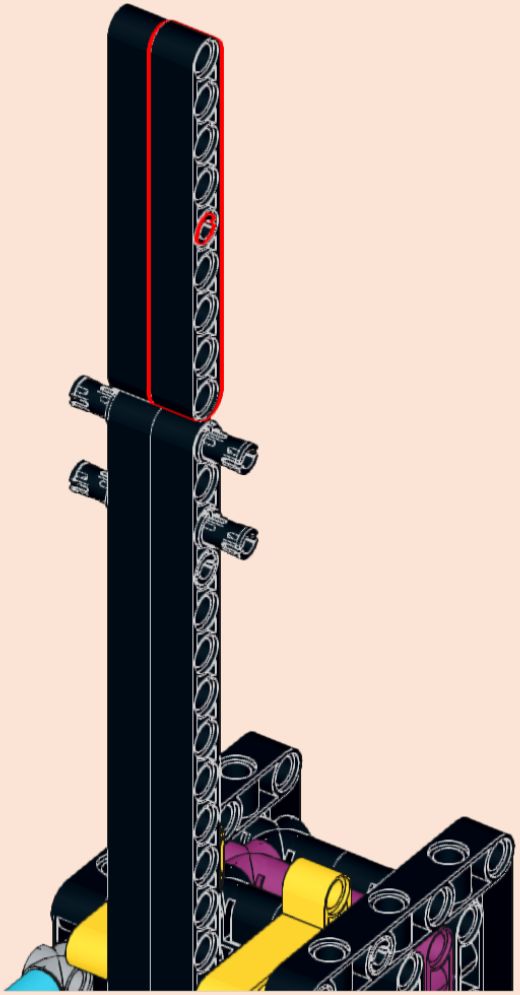
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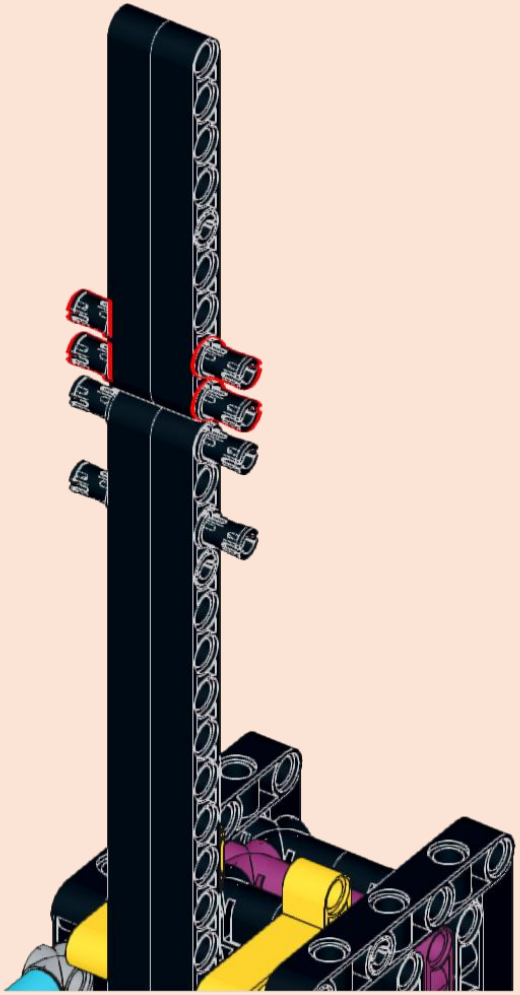
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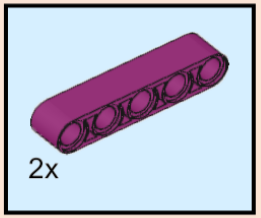


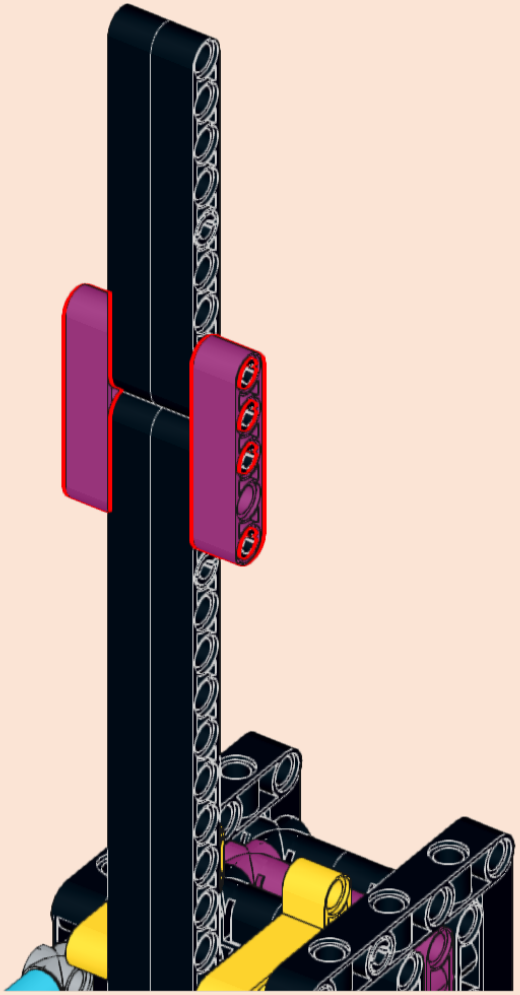
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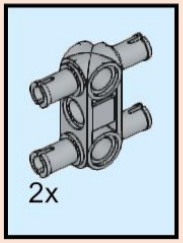


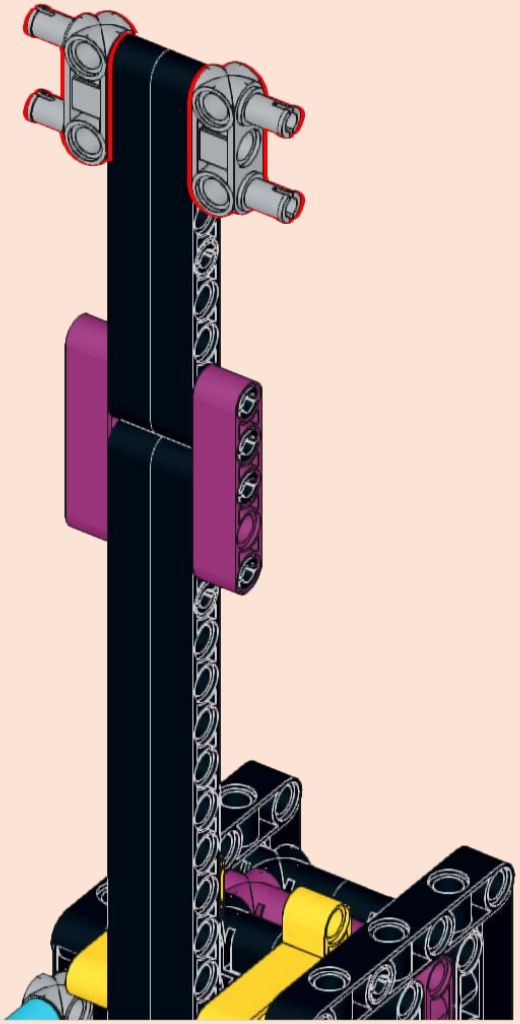
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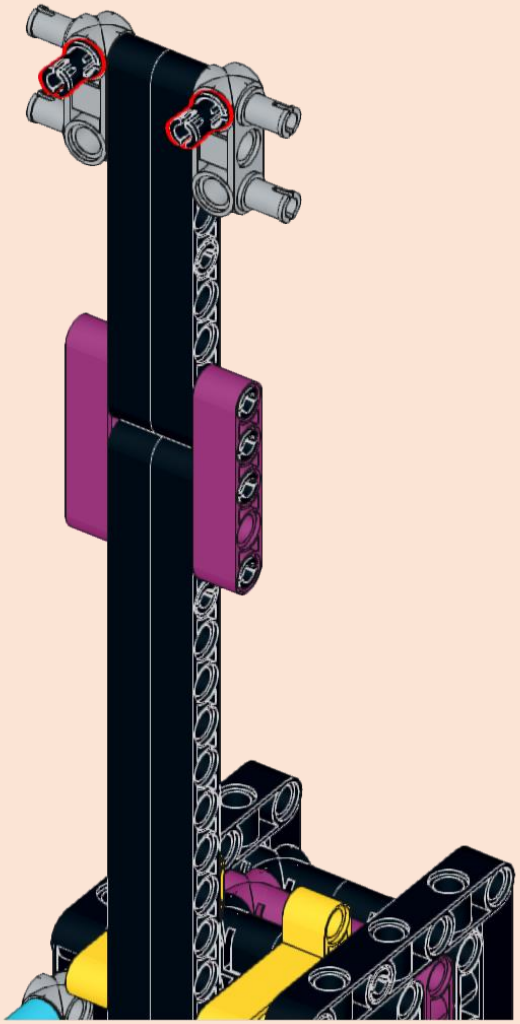
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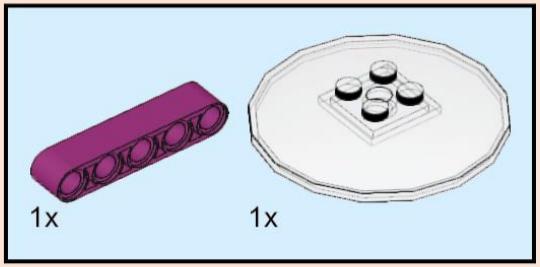


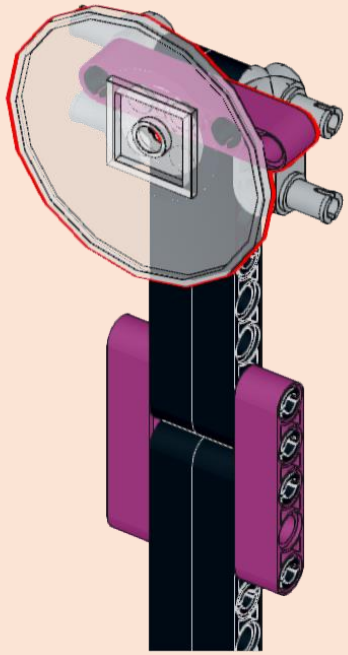
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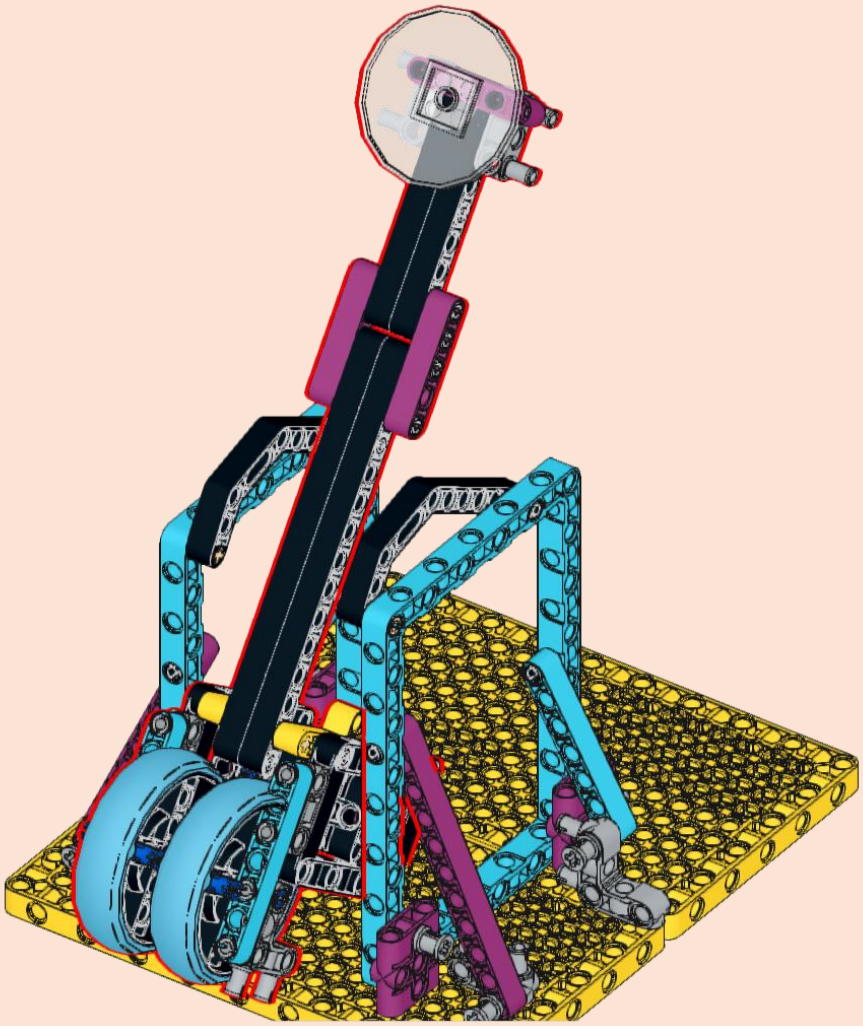


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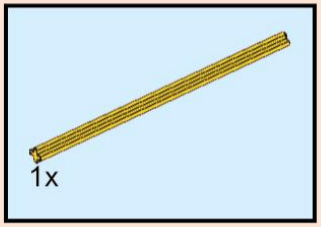


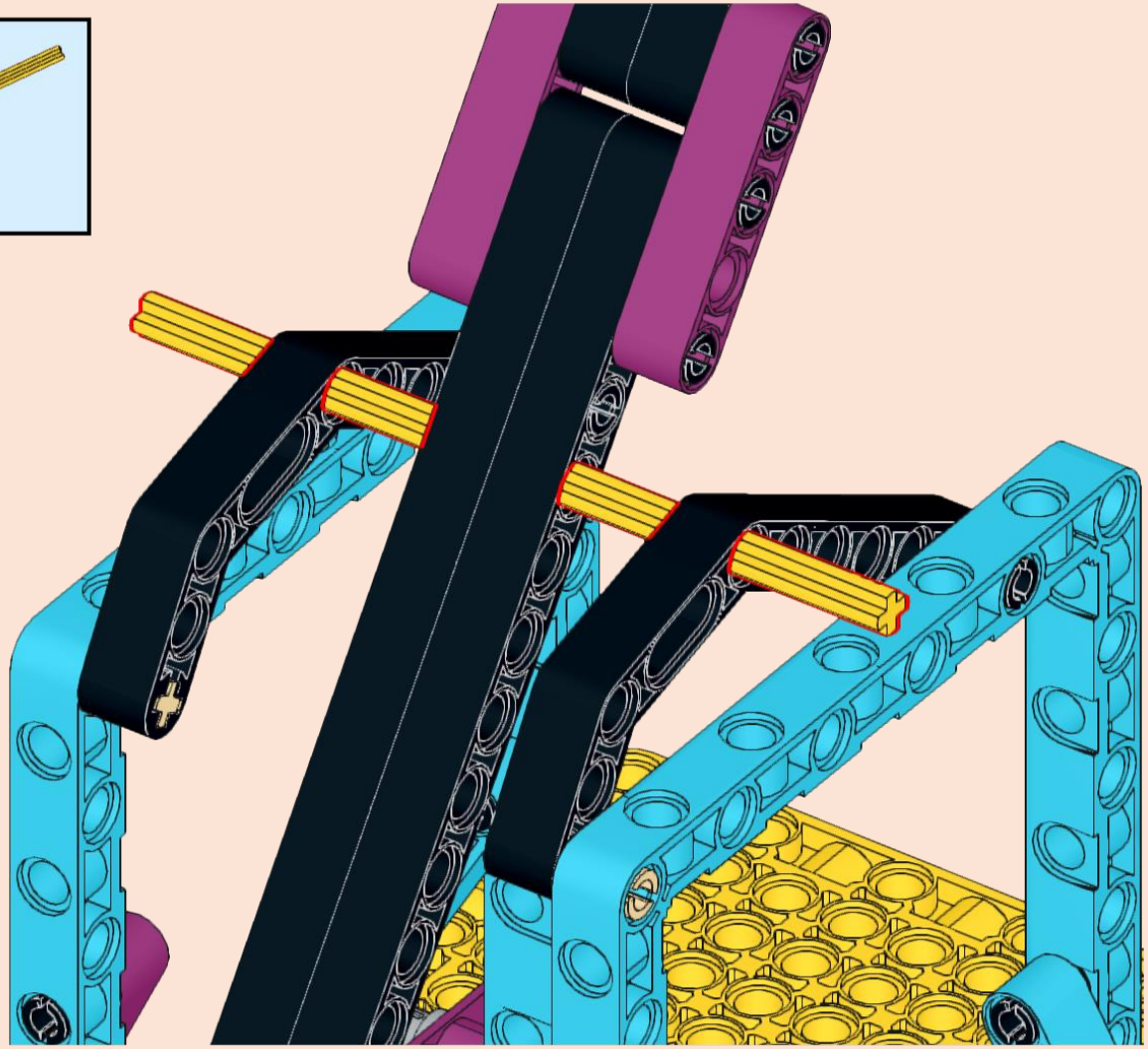


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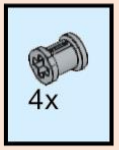


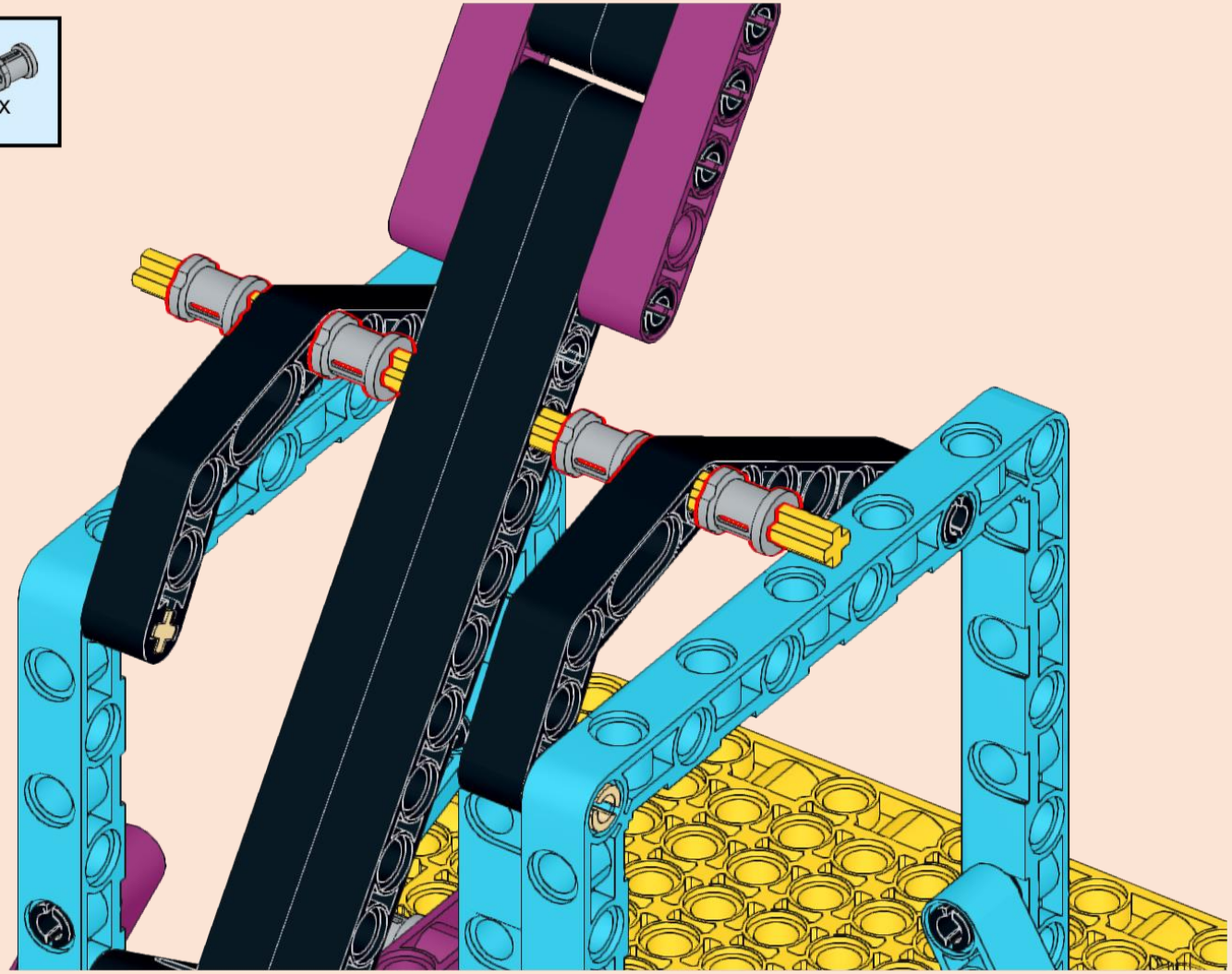
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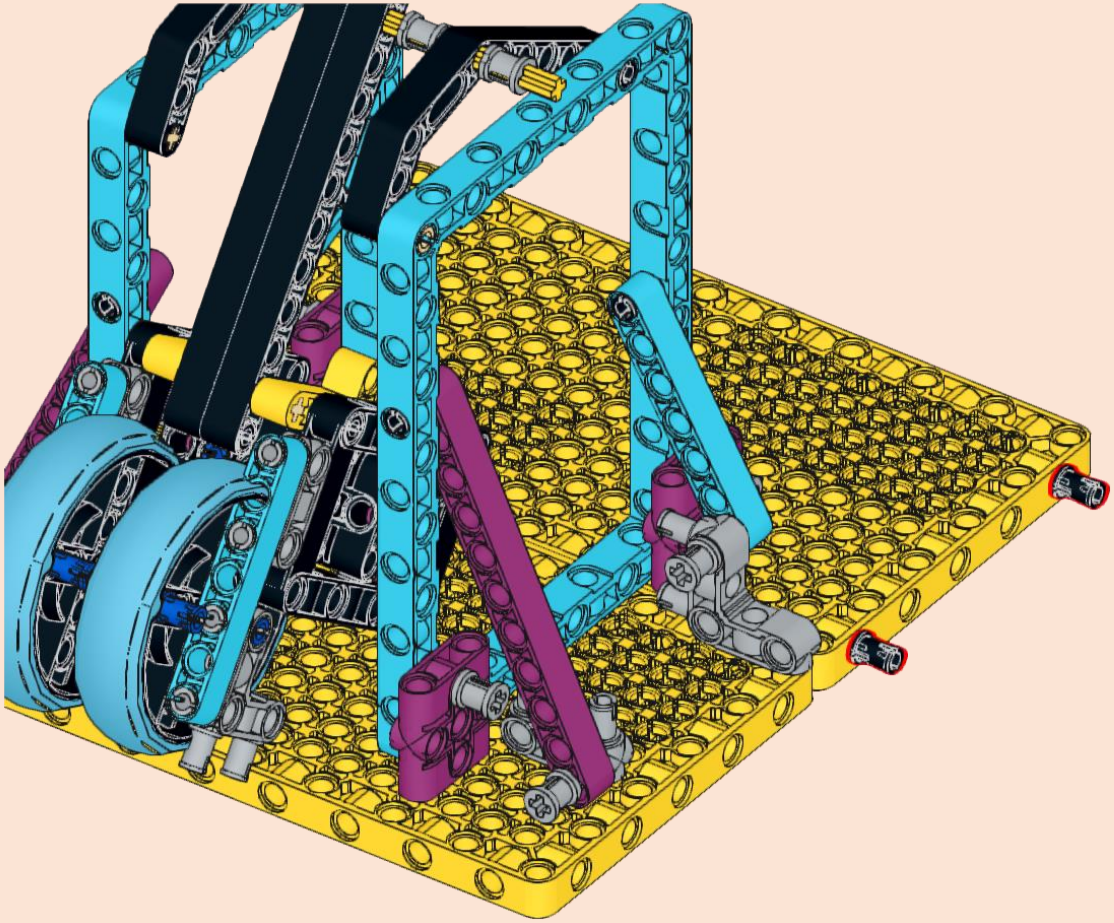




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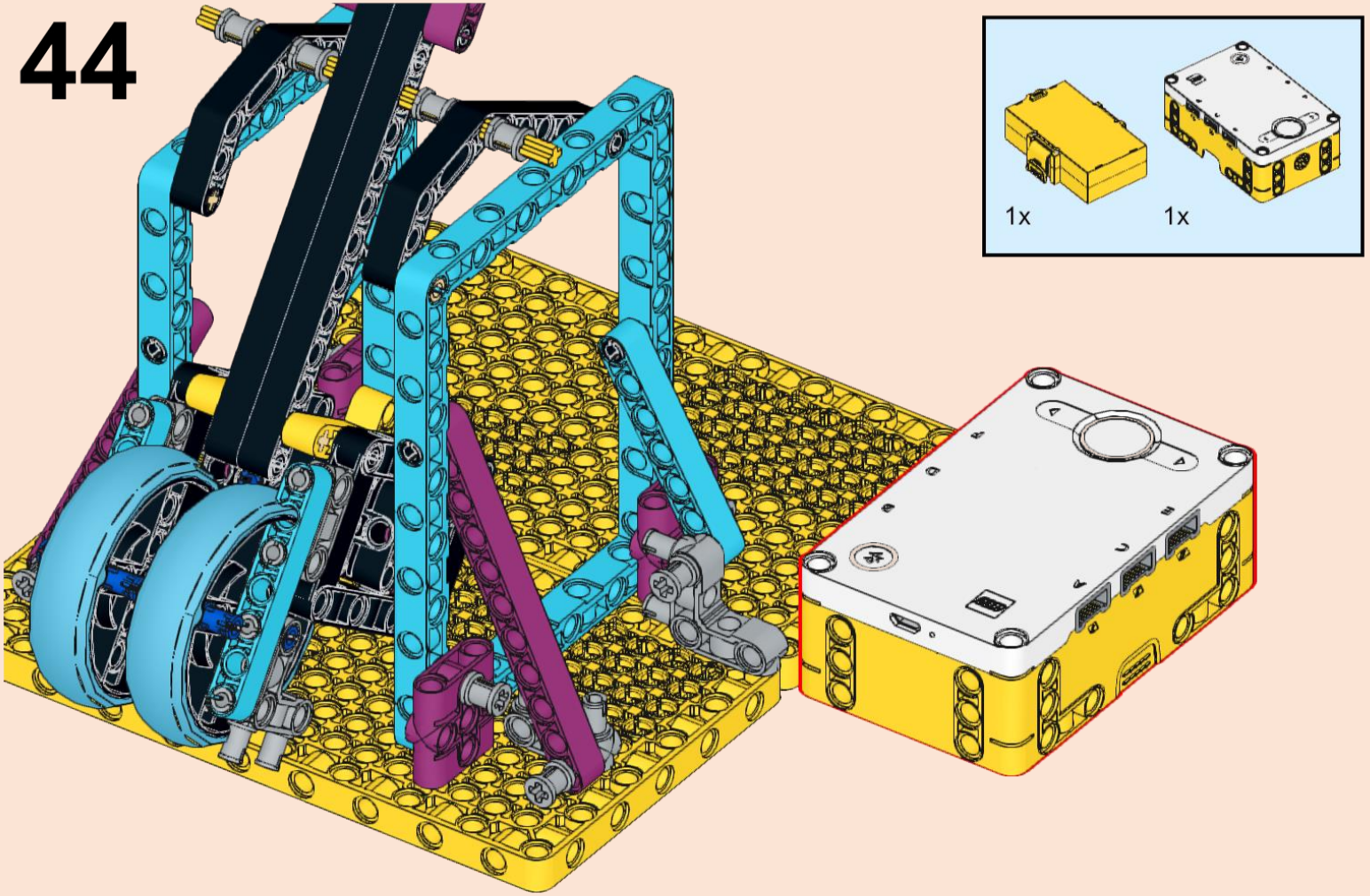


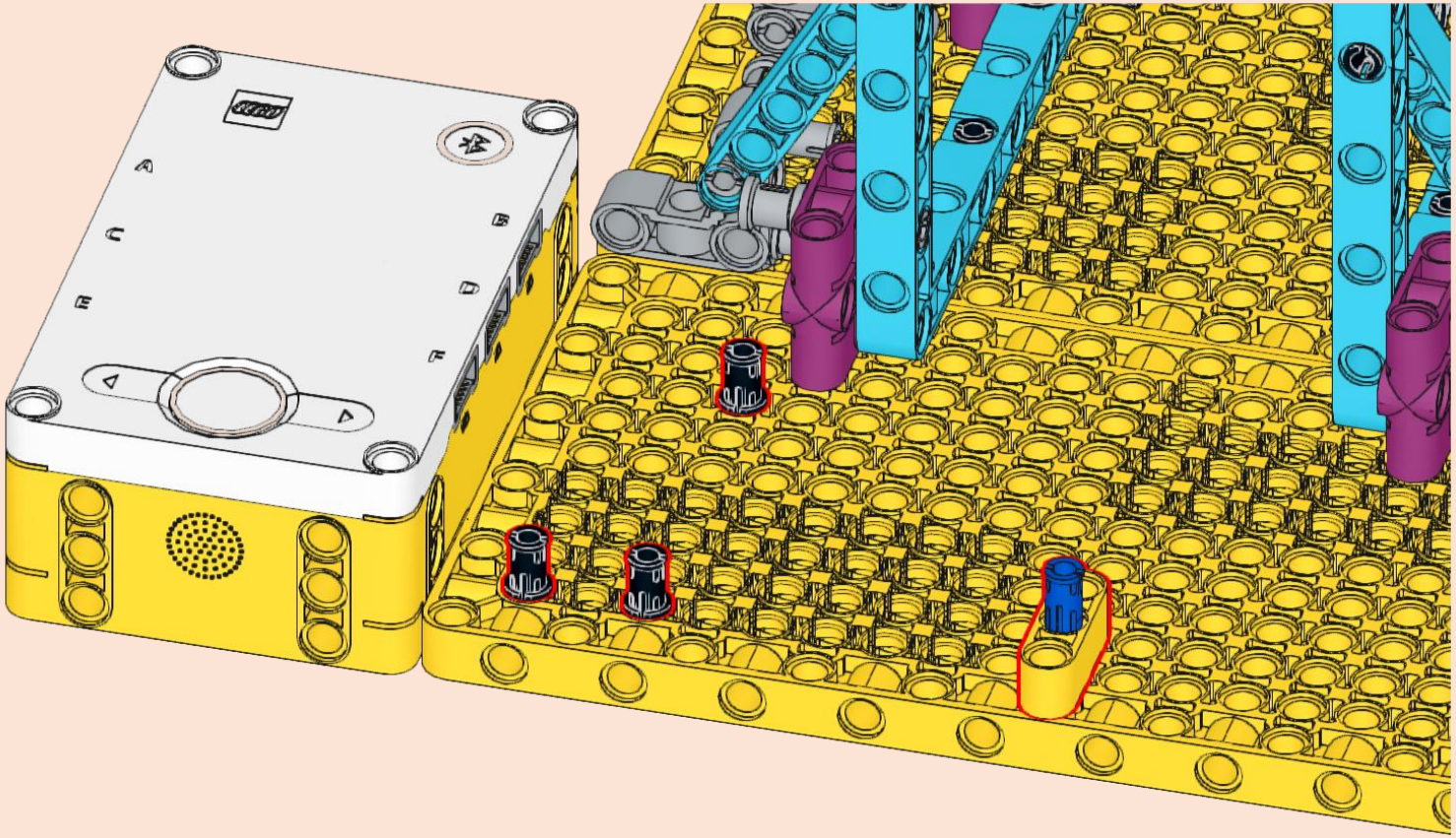




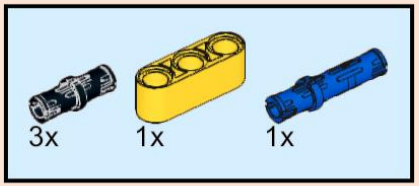


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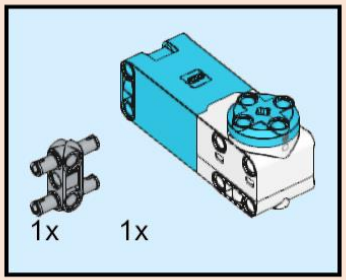

44
1x
1x



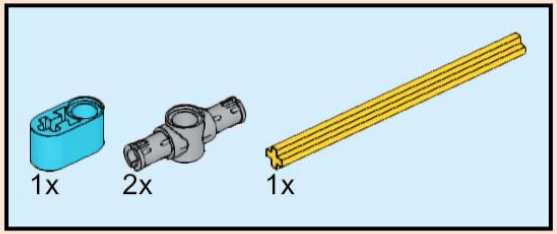
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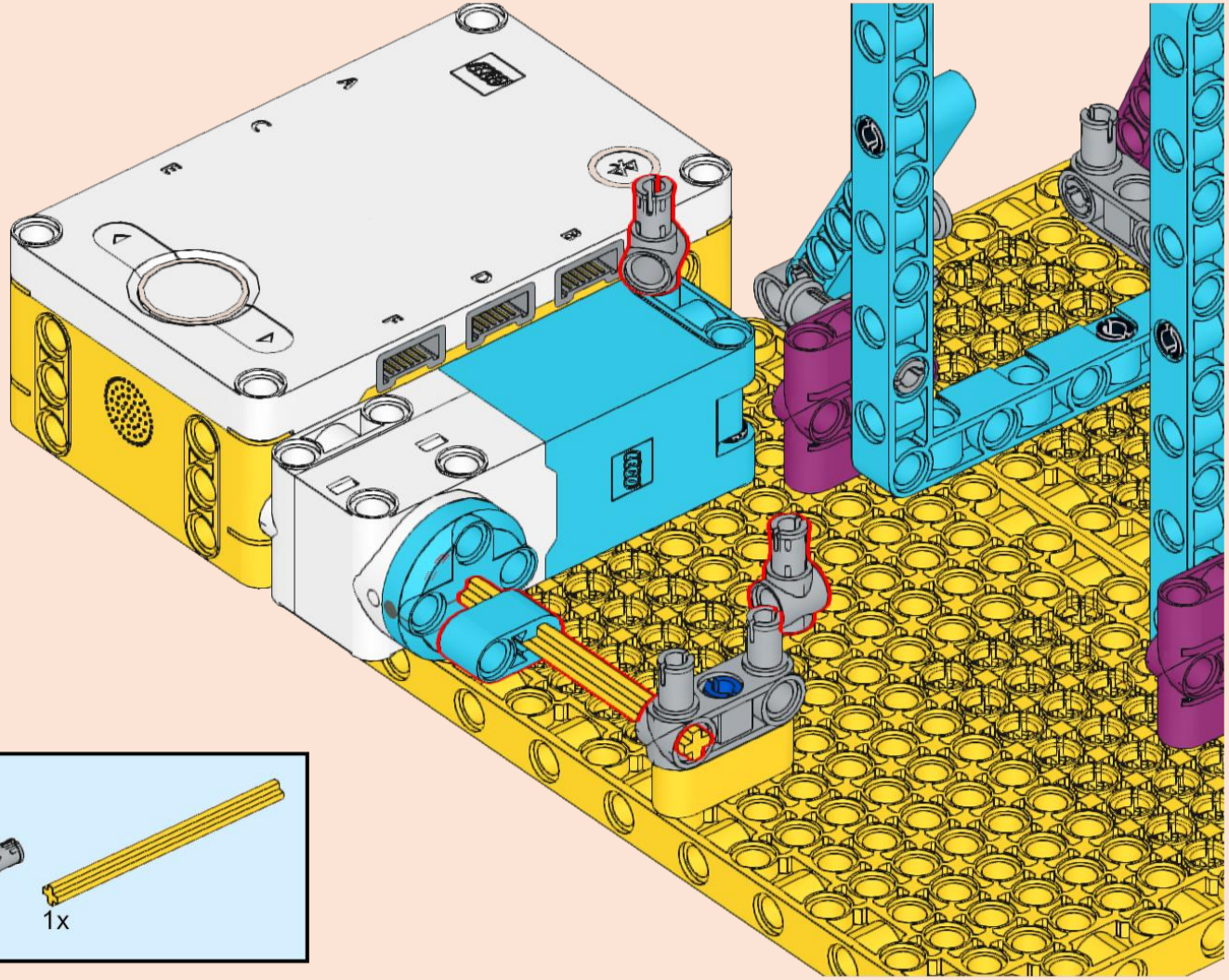


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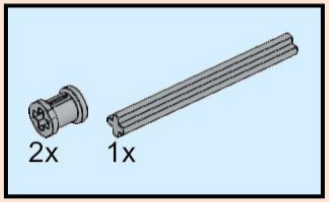


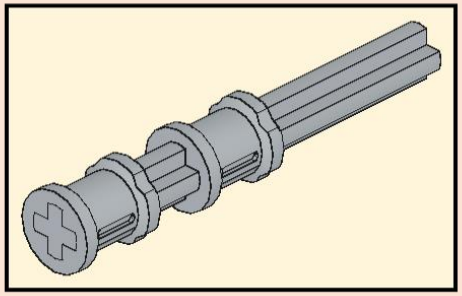
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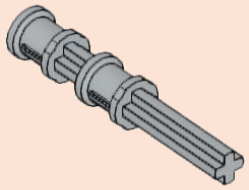


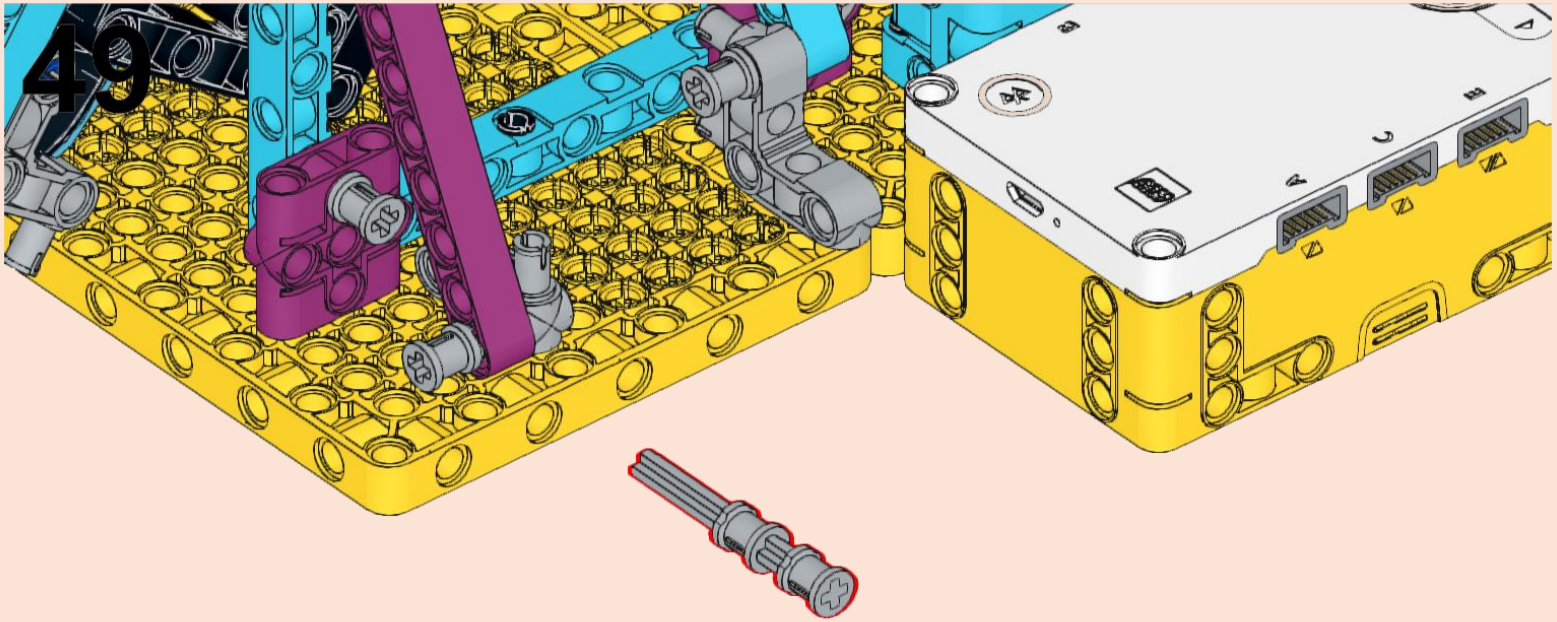


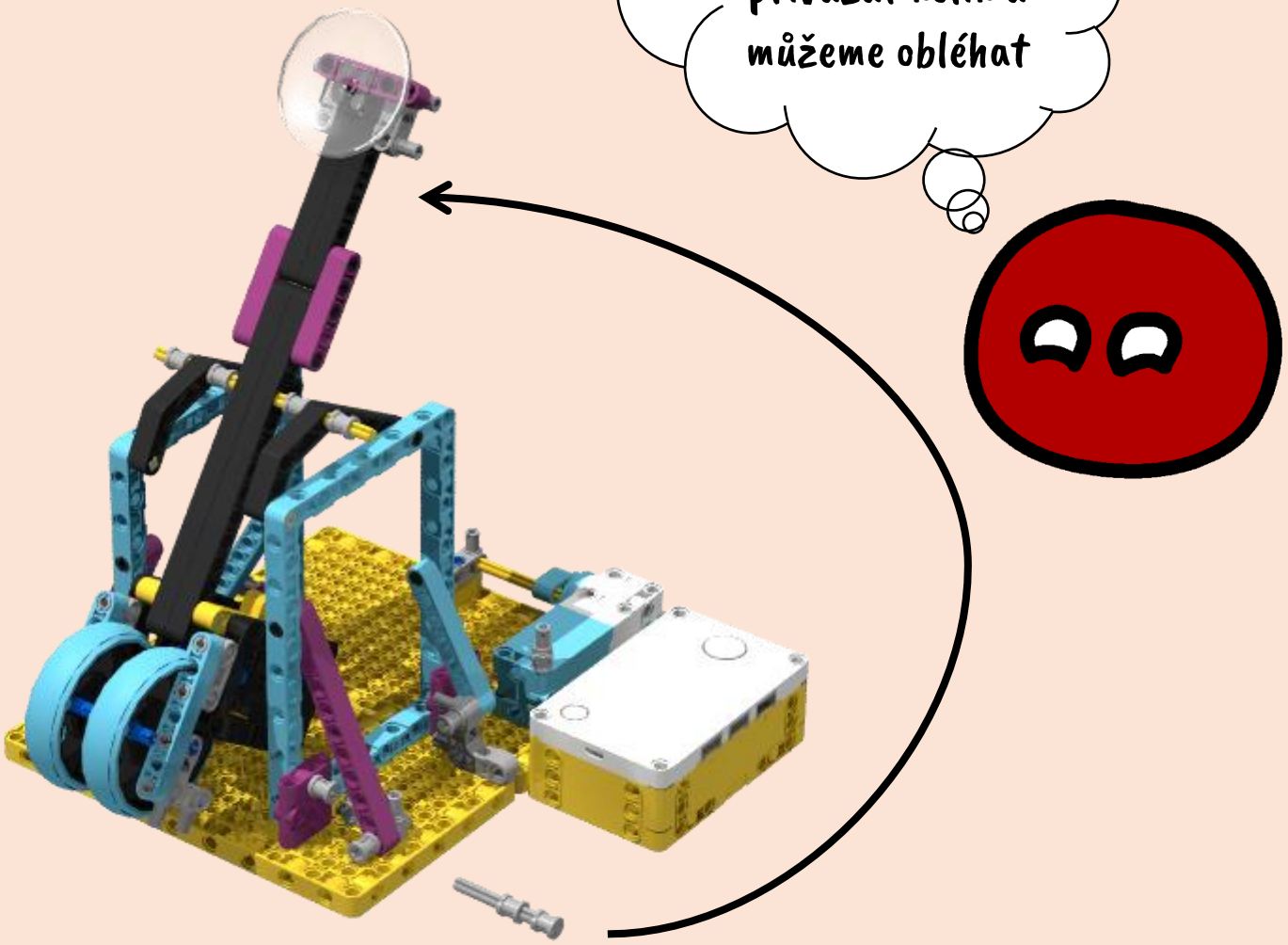
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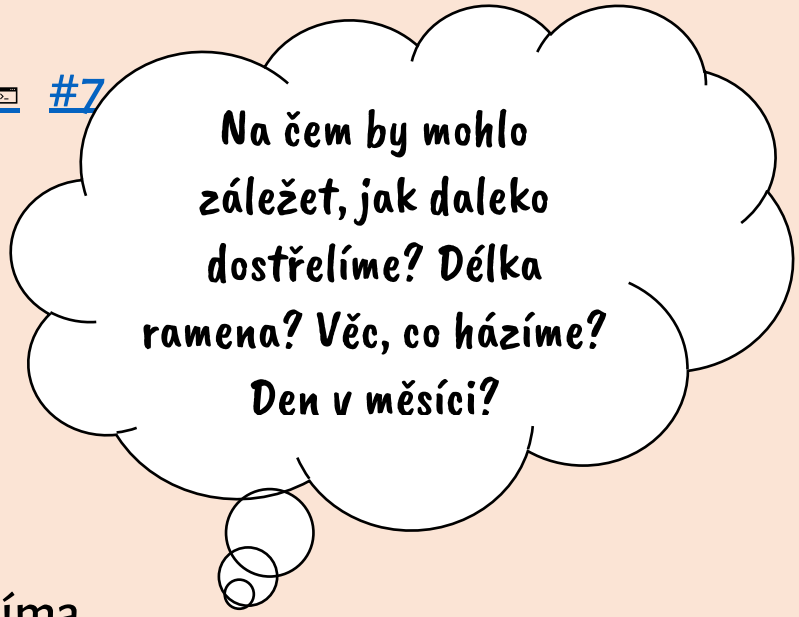

Paráda, už jen přivázat kolík a můžeme obléhat

Jde se programovat!

Výzvy:

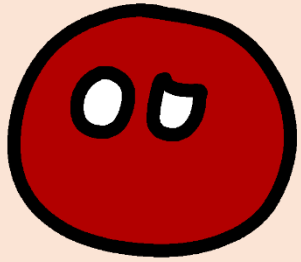
- ☐ Nastavit jednoduché navíjení/odvíjení provázku/nitě na osu s pomocí motoru. (Ovládat se může třeba pomocí držení tlačítek na bricku)
  - Nastavit podle provázku co používáte maximální úhel/pozici motoru, za kterou už se dál nebude moct navíjet. (bezpečnostní pojistka)
  - + Namísto držení tlačítek pro navíjení naprogramovat 2 polohy, (Nataženo – Vypuštěno) do kterých se motor nastaví sám po zmáčknutí tlačítka. (kolík se musí vytahovat ručně)

Čas na testování a úpravy



Výzvy:

- ☐
- ☐ Pokusit se upravit katapult tak, aby vrhnul co nejdál!
  - ○ Zaznamenávejte vzdálenosti a porovnejte s ostatníma.
  - ○ Nejdelší vzdálenost rozdělit na 3 díly a naprogramovat 3 natažení katapultu které dopadnou co nejblíže na určené vzdálenosti.



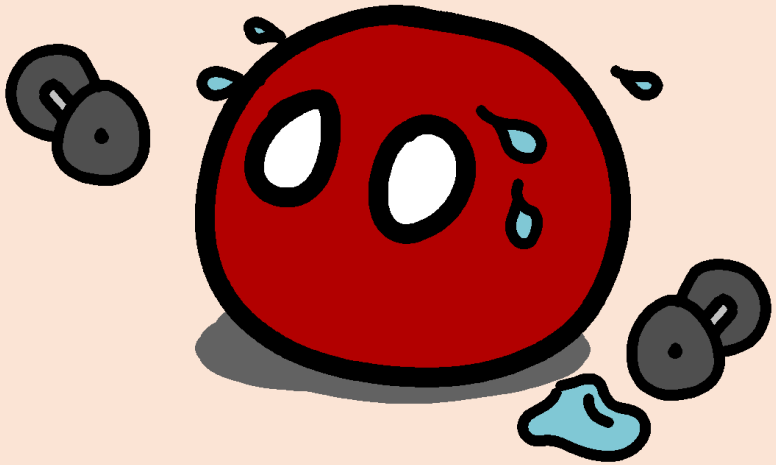
- ☐ + Zkusit upravit navíjející mechanismus, aby navíjel rychleji bez zrychlení motoru.
- ☐ + Přestavět katapult a místo síly gravitace pohánět házecí rameno Lego motory.

Shrnutí a zakončení

- ☐ Porovnejte s ostatníma, kdo má nejlepší katapult? Co ti šlo? Co ti přišlo těžký?
- ☐ Jestli katapult funguje, jak má, co třeba slepit kostky z papíru, postavit věže a překonávat se v pevnější stavbě?

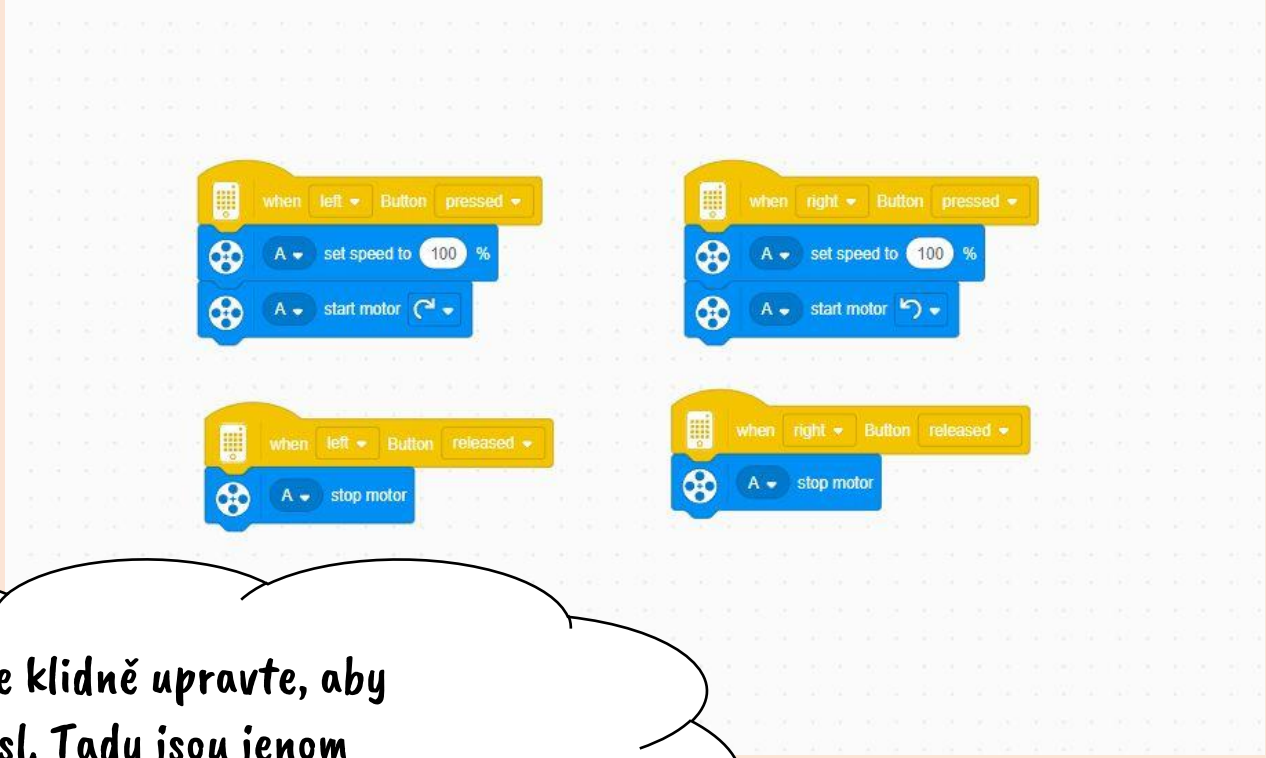






Nápovědy k řešení a přílohy

1- Katapult:



Jakejkoliv kód co tady uvidíte klidně upravte, aby vám vyhovoval a dával smysl. Tady jsou jenom nápady, jak jsem to udělal já. Určitě by moje bloky šly i zjednodušit, klidně se na to vrhněte.

